SAKLOLO: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter highlights the critical challenges currently hindering effective disaster management in Bulacan and CAMANAVA, specifically focusing on the inefficiency of traditional response mechanisms in processing the massive influx of unstructured digital communications during disasters and emergencies. This provides the contextual background and foundation for implementing this computational solution within the local disaster response infrastructure.

Background of the Study

During severe hydrometeorological events, local disaster management offices are consistently overwhelmed by a massive influx of unorganized, high-velocity digital communications from affected citizens. While these localized text feeds contain critical, actionable information regarding immediate rescue needs and structural damage, the sheer volume and unstructured nature of the data render manual processing highly inefficient. To address this information bottleneck, modern computational semantics relies on vector space mechanics to mathematically transform qualitative narrative posts into numeric matrices. As established by Mihunov et al. (2022) mapping unstructured disaster text into a high-dimensional geometric space allows automated systems to systematically extract abstract semantic relationships. This mathematical transformation reduces chaotic crisis communications into distinct spatial coordinates, enabling the systematic and automated categorization of emergency surveillance data.

Building upon this developments, recent computational research has successfully de-

ployed probabilistic generative models to filter, group, and route these high-volume disaster reports in real time. Previous empirical studies demonstrate that algorithms such as Latent Dirichlet Allocation (LDA) effectively manage the linguistic noise, broken syntax, and data sparsity inherent in online text feeds during acute natural hazards. For instance, during the 2023 Hebei Floods, Zang (2026) utilized LDA topic modeling to process unorganized short-text feeds, proving that probabilistic topic distributions can successfully categorize heterogeneous public discourse into actionable, distinct thematic clusters. By treating each text report as a probabilistic mixture of topics rather than relying on rigid string-matching, these models allow crisis managers to bypass information overload and automatically isolate urgent, localized operational needs from general baseline noise.

While these computational frameworks demonstrates profound efficacy in international applications, their deployment is most critical in regions facing compounding, high-frequency environmental crises. The necessity for automated, real-time sensemaking tools becomes acutely apparent when contextualizing the hyper-vulnerable geographic landscape of the Philippine archipelago. In an environment where municipal emergency management offices are repeatedly overwhelmed by simultaneous localized crises, the rapid algorithmic extraction of actionable disaster data is not merely a logistical optimization, but a fundamental requirement for survival and resilience. This operational urgency is directly driven by the country's severe climatic reality.

Disaster-related risks in the Philippines are numerous, as the country is heavily exposed to natural disasters. It is ranked eighth as the most affected globally by extreme weather events, and at least 74% of Filipinos are vulnerable to multiple hazards. Among these hazards, the most catastrophic are tropical cyclones, flooding, storm surges, droughts, earthquakes, and volcanic eruptions. Climate change has caused these events to increase in intensity, as over the past five years the country has experienced destructive typhoons. These disasters have caused economic losses and casualties (World Bank Group, 2023).

Key challenges in implementing Disaster Risk Management (DRM) programs include

institutional fragmentation, lack of coordination, and insufficient implementation capacity. The increasing frequency and severity of disasters further emphasize the need for integrated climate change adaptation and mitigation measures to strengthen the resilience of Local Government Units (LGUs). World Bank Group (2023) highlights the need to clarify the roles, responsibilities, and improved resource allocation between the National Government and LGUs to ensure the effective implementation of disaster preparedness, response, and recovery interventions.

While social media is utilized for disaster response, the way it is utilized, and in what context, might point to a more nuanced landscape regarding its limitations and the nature of its usage by residents in a certain area. A literature review by Inoferio et al. (2025) found that social media disaster response is not uniformly developed for all aspects, with man-made disasters not as developed compared to natural disasters in utilization, while COVID-19 and vegetation fires have not been explored as a means for disaster response.

Its utilization was heavy during response and recovery and minimal during preparation. Misinformation, noise, and infrastructure limitations are a concern. It is highly effective in coordinating recovery and providing updates, provided they are filtered between the informative and the controversial Inoferio et al. (2025)—possibly implying that sophisticated ways to process information and deliver actionable insights to stakeholders could help communities become more resilient and responsive in conducting disaster response.

To bridge this gap, this study proposes *SAKLOLO: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA*. This framework introduces a novel hybrid architecture (**System A**) that combines Count Vectorization, Latent Dirichlet Allocation (LDA), and pre-trained XLM-RoBERTa-based contextual representations into a Joint Matrix. This approach preserves mathematical interpretability while achieving high classification accuracy on chaotic, mixed-syntax Taglish text. Furthermore, the study conducts a rigorous comparative benchmark of this lightweight system against a state-of-the-art open-weight dense transformer (**System B**: Qwen3-1.7B

-Instruct-Q4_K_M) operating under few-shot prompting conditions, which is also expected to yield a joint matrix.

The study is focuses on modern-day climate risks and high social media penetration in the Philippines, as understanding the digital sentiments and operational needs of vulnerable communities is crucial. This automated processing allows for a tailored response to specific community needs rather than a generalized strategy, enabling informed resource allocation of goods, effort, and time. By benchmarking deterministic and probabilistic machine learning models as a relative reference against generative AI LLMs, the study ensures that disaster management offices can focus on their core competencies effective and well aligned information processing and dissemination. Rather than manually parsing through an overwhelming influx of digital noise.

Statement of the Problem

This study evaluates and compares the performance of a hybrid joint matrix pipeline as the baseline (**System A**) against a state-of-the-art large language model joint matrix output (Qwen3-1.7B-Instruct-Q4_K_M) in processing unstructured disaster-related text to generate actionable sentiment classifications and topic modeling for the Bulacan and CAMANAVA regions. Specifically, it seeks to answer the following questions:

1. Under Bivariate Analysis, to what extent do the topics from the joint matrix of both systems show statistically significant association with sentiment classifications as measured by:

    a. Chi-Square;

    b. Fisher's Exact Test (Contingent on Chi-Square's assumptions not holding).

2. Is there a statistically significant operational difference between the outputs of **System A** and **System B** as measured through:

a. McNemar's Test (with continuity correction).

Significance of the Study

SAKLOLO: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA is a computational information system for disaster response decision-making. It evaluates how unstructured information in disaster situations can be optimally processed using a Hybrid LDA + XLM-RoBERTa Joint Matrix compared against a `Qwen3-1.7B-Instruct-Q4_K_M` joint matrix pipeline. By extracting meaningful information—including semantic topics and contextual sentiment—from unstructured emergency feeds, the study establishes a mathematically rigorous framework to understand a disaster and respond to it effectively. This study benefits several individuals and institutions:

Bulacan and CAMANAVA Disaster Response Offices. The study is intends to aid disaster response offices in organizing and structuring the unorganized disaster-related information they receive through social media posts. By determining whether a lightweight hybrid architecture or a open-weight Large Language Model is more operationally viable for joint latent topic modeling and its respective sentiment, DRRMO employees are provided with an optimized method to see what disaster-related posts exist and what the sentiments regarding these posts are as received by the concerned communities. This enables personnel to better understand localized field problems and provides them with empirical insight when planning response strategies and disseminating information.

Local Government Units (LGUs). The study assists LGUs in understand- ing emergency situations occurring in their specific jurisdictions by automatically structuring localized reports. Identifying the most urgent issues through sentiment and latent topic analysis helps LGUs gain insights into the current state and emotions their respective citizens have during natural disasters. This aids in assessing the disaster situation locally and assist LGUs

in making decisions that are more aligned, and in conducting messaging that takes into consideration how their citizens talk about sets of words that have topics, and what their sentiment is around those co-occurring sets of words in social media posts.

Emergency Responders. The study is beneficial to emergency responders by providing disaster-related information in a more organized format. As opposed to reviewing social media posts as isolated and unorganized streams of information, responders can evaluate co-occurring words, their sentiments, and how their respective joint matrices relate to each other. This organized intelligence enables emergency responders to make highly informed, emotionally intelligent, and community-aligned messaging that aids disaster response.

Researchers and Future Developers. The study acts as a mathematical and computational foundation for those interested in natural language processing within high-stress, low-resource languages (Taglish). Specifically, the novel joint matrix approach (combining LDA topic probabilities with XLM-RoBERTa contextual embeddings) provides reproducible evidence on hybrid architecture performance versus dense large language models. The integration of Bayesian inference, information theory, and comparative deep learning establishes a robust baseline for future researchers pursuing text optimization in Philippine disaster management.

Alignment with Sustainable Development Goals

This study primarily aligns with United Nations (2020) Sustainable Development Goal 11 (Sustainable Cities and Communities), particularly Target 11.5, which emphasizes reducing the adverse impacts of disasters on human lives and communities. Through the development of SAKLOLO, the study seeks to contribute to this goal by providing an analytical approach to processing and organizing disaster-related social media information. By identifying prominent discussion topics and their associated sentiment classifications, SAKLOLO aims to provide disaster response offices and other concerned stakeholders with

additional information for understanding community concerns, identifying topics that warrant further investigation, and informing information dissemination and response planning. The resulting topic-sentiment relationships may also provide an analytical basis for considering which reported concerns require greater attention when planning the allocation of resources.

The study also aligns with United Nations (2020) Sustainable Development Goal 13 (Climate Action), particularly Target 13.1, which emphasizes strengthening resilience and adaptive capacity to climate-related hazards and natural disasters. SAKLOLO supports this alignment through the automated analysis of social media reports concerning typhoons, flooding, and other hydrometeorological events. Through its Decision Support System component, the study presents organized topic and sentiment information that may provide additional context for understanding public concerns during and following disaster events. By making these patterns more accessible for examination, SAKLOLO seeks to contribute an additional analytical resource for disaster-related assessment and planning.

Secondarily, the study contributes to United Nations (2020) Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) through the development and comparative evaluation of hybrid computational and artificial intelligence approaches for processing multilingual emergency-related information. Specifically, the study examines the applicability of these approaches to Tagalog, English, and Taglish social media posts and comments within the Philippine context. Through this evaluation, the research explores the potential of combining computational and artificial intelligence techniques to analyze unstructured, multilingual information in the context of disaster-related discussions.

Scope and Delimitation

This study focuses on the comparative quantitative evaluation of two computational systems designed to perform topic modeling and sentiment analysis on unstructured disaster-related reports. **System A** is a hybrid pipeline utilizing Count Vectorization, Latent Dirichlet

Allocation (LDA), and XLM-RoBERTa contextual representations combined into a Joint Matrix . **System B** utilizes `Qwen3-1.7B-Instruct-Q4_K_M`, an open-weight large language model, operating under few-shot prompting conditions, the two systems will be compared in a same measurement comparative analysis.

The geographic scope of the study covers the province of Bulacan and the CAMANAVA region (Caloocan, Malabon, Navotas, and Valenzuela). The temporal scope of the corpus spans a five-year period from 2020 to 2025. Data collection is strictly limited to publicly accessible Facebook posts and comments written in Tagalog, English, and Taglish that meet the established inclusion criteria regarding natural disasters (e.g., typhoons, flooding).

The evaluation framework is strictly delimited to specific mathematical and statistical metrics. Joint Matrix analysis is to be conducted to analytically test the categorical associations of words to their sentiment using Chi-square, and a contingent Fisher's Exact Test. The operational utility of the systems is evaluated through a purposive stakeholder assessment using McNemar's Test.

The study does not involve real-time integration into actual DRRM command centers, **nor does it include the physical dispatch of rescue teams**. It performs social media post backend aggregation with respect to the Data Privacy Act of 2012; all sampled posts and comments are anonymized and protect the identity of the Facebook users. **System A** features a pre-trained XLM-RoBERTa (`cardiffnlp/twitter-XLM-RoBERTa-base-sentiment`) that features approximately one million Tagalog social media posts as part of its training dataset alongside vast numbers of other languages.

Regarding system implementation, the backend computational pipeline is encapsulated within a **localized analytical web dashboard** developed using a lightweight Python-based web framework (e.g., Streamlit). This dashboard is strictly delimited to an partially online, local-host analytical tool designed for the researchers and the expert stakeholder panel. This study does not produce a publicly hosted, enterprise-grade web application, nor does it include real-time cloud deployment, multi-user authentication, or live integration into

actual LGU command centers.

CHAPTER II

THEORETICAL FRAMEWORK

This chapter outlines the foundations supporting the development of *SAKLOLO: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA*. It is grounded in Vector Space Semantics, the Distributional Hypothesis, and Bayesian Generative Theory, justifying the probabilistic extraction of semantic structures from unstructured disaster text. The literature review discusses Count Vectorization mechanics, contextual language representations (XLM-RoBERTa), generative large language models (LLMs), and Latent Dirichlet Allocation (LDA) applications in crisis environments. A synthesis highlights gaps in processing mixed-language (Taglish) emergency feeds and guides the proposed mathematical joint matrix model.

Related Theories

The conceptual foundation of this research is grounded in Sensemaking Theory, which asserts that in digital crisis environments, offloading the initial chaotic, high-velocity data to automated systems allows human responders to bypass information overload and focus directly on high-level resource allocation (Pezanowski et al., 2022). To mathematically achieve this automated sensemaking, the computational framework relies on the Distributional Hypothesis in computational semantics. Balkir et al. (2016) explain that this hypothesis assumes words occurring in similar contexts inherently share similar semantic meanings. This provides the theoretical justification for mapping text into vector spaces. It theorizes that even highly disorganized human crisis communication contains latent geometric patterns, allowing related disaster concepts to naturally cluster together when analyzed

algebraically.

Complementing this geometric assumption is the statistical principle of rxchangeability, rooted in De Finetti's theorem. Blei et al. (2003) utilized this theory to establish that when modeling text data, the joint probability of a sequence of terms is invariant to their specific grammatical order. This provides the core theoretical justification for the "bag-of-words" approach used in probabilistic topic modeling. Because the theory assumes that documents are the end result of a probabilistic generative process rather than rigid grammatical structures, the proposed system is mathematically validated in ignoring the broken syntax and informal structures typical of emergency posts, focusing purely on the statistical distribution of variables to uncover hidden situational categories.

Review of Related Literature

Computational Frameworks in International Disaster Literature

The theoretical foundational framework for processing unstructured crisis communications relies on computational linear algebra, vector space modeling, and probabilistic generative models. According to Mihunov et al. (2022), vector space mechanics represent qualitative narrative posts as numeric constructs in high-dimensional matrices to enable systematic computation during disasters. Within this space, a document–term matrix captures documents across unique vocabulary tokens, where raw text is mathematically transformed using *count vectorization* to systematically extract critical crisis keywords for downstream processing.

Complementing these deterministic vector decompositions, probabilistic frameworks model the underlying semantic structure of text. Blei et al. (2003) introduced Latent Dirichlet Allocation (LDA) as a Bayesian generative probabilistic model that views each document as a multinomial distribution over latent topics, which are themselves multinomial distributions over vocabulary words governed by Dirichlet priors. Expanding on this

by bridging probabilistic topic models with deep learning, Zang (2026) highlights the integration of LDA with contextual transformer architectures (such as BERT). Inferring hidden topic assignment variables alongside deep contextual embeddings from observed social text feeds allows for highly accurate, granular identification and tracking of emergent situational themes as rapid disaster events unfold.

Social Media Utilization in Philippine Disaster Risk Reduction

A literature review on the role of social media in disaster response in the Philippines found that it was heavily employed in response to typhoons and earthquakes, but other disasters such as COVID-19, floods, and vegetation fires are not fully explored. The review indicated that the usage of social media was heavy during response and recovery but minimal during preparation. Because misinformation, noise, and infrastructure limitations remain a concern for its utilization as a reliable disaster response tool, a hybrid computational approach is recommended. Social media was found to be highly effective in coordinating recovery and updates, provided the data is computationally filtered to separate the informative from the controversial (Inoferio et al., 2025).

The Philippines has four thematic areas of Risk Reduction and Management. Disaster Prevention and Mitigation is aimed at strengthening community resilience by minimizing exposure to risks and increasing recovery capacity. Disaster Preparedness involves strengthening community and institutional capacity via collaborative policies and systems. Disaster Response focuses on the reduction of preventable casualties and ensures the basic needs of communities are met and restored rapidly. Finally, Disaster Rehabilitation and Recovery deals with the restoration of livelihoods, economic opportunities, and essential infrastructure that supports the physical and psychological rehabilitation of those affected (Inoferio et al., 2025).

The North Manila Bay Delta consists of the downstream part of the rivers of the Pampanga River Basin, including the coastal area. The wider basin has a catchment area of

$10,000\ \text{km}^2$ and is roughly divided into three river systems. These are the Pampanga main river system (about $8,000\ \text{km}^2$), the Angat River system (about $1,085\ \text{km}^2$), and the Pasac river system (about $1,371\ \text{km}^2$), which originates from Mt. Pinatubo and is connected to the Pampanga River by the Bebe-San Esteban Cut-off Channel (Royal HaskoningDHV, 2026).

This delta covers almost $3,000\ \text{km}^2$ of the southern central valley of Luzon, stretching from Pampanga to Bataan, Bulacan, and the CAMANAVA area of Metro Manila. The plains are so flat and close to sea level that the 1-meter elevation line extends 10 to 20 km inland, so even a small rise in relative sea level translates to a large encroachment of the sea. The region is also a major rice, sugar, and fish producer, and groundwater is its main source of fresh water for households, industries, fishponds, and farms (Rodolfo et al., 2003).

Linguistic Context and Geographic Scope Selection

There are four main Tagalog dialects: Northern Tagalog Dialects of Bulacan (Bulacan & Nueva Ecija) and Bataan (Bataan & Zambales), Central Tagalog Dialects of Laguna (Laguna & Rizal) and Metro Manila, Southern Tagalog Dialects of Batangas (Cavite & Batangas), Lubang (Mindoro), and Tayabas (Aurora, Camarines Norte, Camarines Sur, & Quezon), and lastly Tagalog Dialects of Marinduque (Adriano, 2023).

The North Manila Bay Delta shares common hydrometeorological traits of natural disaster frequencies and structural instances. It consists of minor parts of Bataan, Pampanga, Bulacan, and CAMANAVA. Bataan is omitted from this study due to minor inclusions on the delta; while Pampanga is omitted due to not being included in major Tagalog dialects and having a different linguistic profile among its speakers. That leaves Bulacan and CAMANAVA as the geographic scope of the study due to sharing the same hydrometeorological profile and relatively similar linguistic profile of Tagalog (Northern and Central dialect), English, and Taglish.

Review of Related Studies

Global Evaluations of Text Processing in Disaster Management

Empirical implementations across Asian and global disaster contexts demonstrate the practicality of combining probabilistic topic modeling with transformer-based deep learning to process high-velocity unstructured social text feeds. Zang (2026) utilized the LDA topic modeling algorithm alongside prompt-BERT models to process unorganized short-text feeds during the 2023 Hebei Floods in China, proving that hybrid architectures effectively categorize heterogeneous social discourse into actionable thematic clusters for crisis managers.

Evaluating short-text social streams during major hydrometeorological events further highlights the necessity of strict statistical metric validation. Anggara et al. (2025) applied LDA topic modeling to Twitter data during the 2024 West Sumatra flash floods, demonstrating that evaluating topic coherence scores (C_U) across various topic ranks (K) isolates the optimal matrix structure, ensuring distinct and semantically meaningful categories without topic overlap.

In Western disaster management contexts, feature extraction and topic modeling have proven vital for real-time situational awareness. Mihunov et al. (2022) used LDA to analyze disaster impact feeds during Hurricane Harvey, successfully isolating 9 high-priority target topic vectors out of 24 computed themes to directly map physical infrastructure damage, power grid outages, and localized civilian entrapment locations.

Further demonstrating the efficacy of localized text extraction during severe flooding in Southeast Asia, Choirul Rahmadan et al. (2020) investigated the Jakarta flash flood domain by applying Latent Dirichlet Allocation (LDA) to high-velocity Twitter streams. Their research successfully clustered unstructured text feeds by localized impact, utilizing the algorithm to map explicit target tokens that detailed specific flooded neighborhoods, severe water depth thresholds, and immediate rescue dispatch needs. Because the hydrology and

social media usage in Jakarta closely mirror the urban–rural challenges found in Philippine municipalities, this study proves that probabilistic generative models can segment baseline social media noise from urgent environmental disaster events to directly support local municipal response.

Local Applications of Topic Modeling and Sentiment Analysis in Crisis Studies

In a local context, Visayan scholars from the University of San Carlos of Cebu and Cebu Technological University utilized Latent Dirichlet Allocation (LDA) and the Hierarchical Dirichlet Process (HDP) to model a qualitative risk assessment for disaster risk reduction response. They attained qualitative data via the disaster risk reduction suggestions of Malasakit, a crowd-sourced disaster initiative across barangays focusing on how to service vulnerable populations (Gorro et al., 2021).

Before conducting the LDA, they performed data cleaning to remove special characters, stop words, and strings under four letters. They then vectorized their entries as a computational prerequisite. For LDA, they measured perplexity scores to determine how stable the machine learning model was in the face of uncertainty, where a lower score indicated better performance. After fitting the models, they discovered the general structural trends: solid waste management, relief and emergency planning, and early warning systems were the most heavily weighted topic sets (Gorro et al., 2021). This study provides a quantitative approach to determining the optimal number of topics based on perplexity scores, which is a valuable methodological addition to local studies.

Another local study utilized LDA in the context of the COVID-19 pandemic in Cagayan Valley, placing a specific emphasis on sentiment analysis. The study utilized Facebook posts from government agencies between March and June 2020, applying standard preprocessing protocols. Using a Java package called Mallet, they fed the data into an LDA model to extract core topics. The generated topic sets went through labeling to identify common themes with the help of the Municipal Disaster Risk Reduction and Management Office

(MDRRMO) of Saguday, Quirino, utilizing six annotated tests judged by multiple public and private sector experts. The study identified five overarching themes capturing public sentiment: COVID-19 testing, relief operations, contact tracing, government programs, and community quarantine (Pagudpud, 2020).

A separate study conducted by Silliman University researchers in Dumaguete City sourced qualitative data from Twitter (X) to conduct topic modeling and document clustering regarding localized sentiments. Utilizing the Twitter4j API, the program gathered 99,942 tweets between 2015 and 2016. After preprocessing the mixed-language texts (Tagalog and English) utilizing string-searching and dictionary-matching algorithms (Aho–Corasick and Damerau–Levenshtein Distance), 22,633 valid tweets remained for training and testing (Montenegro et al., 2018). The researchers utilized Gibbs Sampling to incorporate observed data into the LDA sampling process. Finally, they conducted sentiment analysis via Support Vector Machines (SVM) to classify the texts into positive, negative, or neutral sentiments. While the SVM approach proved functional, it highlights the methodological progression toward modern hybrid architectures and joint matrices that provide deeper contextual understanding than classical SVMs (Montenegro et al., 2018).

Related Undergraduate Theses in Bulacan State University

Bulacan State University has been instrumental in the development of technological applications related to disaster management through various undergraduate studies. An example of this is the study entitled *LokaMatika: Two-Way Location Sharing and Automated Emergency Response Systems with Trilateration and Kinematics* (Manalo et al., 2024). This study designed a mobile application utilizing Dart and Flutter that allows two-way location sharing to help users report and track the locations of vulnerable individuals during emergencies. The researchers utilized ISO/IEC 25010 to evaluate software quality with 243 respondents.

Another related study is *READI: A Modern Way of Saving Lives Using Mobile Appli-*

cation: Real-Time Emergency Alerts and Disaster Information (De Guzman et al., 2019). The researchers developed a mobile application providing real-time emergency alerts and awareness materials to the public, emphasizing the importance of mobile technology in improving communication during emergencies.

While both studies successfully provided a technological basis for disaster management, they focused strictly on the manual emergency response and communication aspects. Neither study attempted to implement natural language processing pipelines, such as contextual Joint Matrices or generative AI Large Language Models (LLMs), to automatically analyze, cluster, and score sentiments on the unstructured text data inherent to disaster communications.

Review of Related Systems

Integrating large unstructured qualitative data from social networks to make informed choices and aid in decision making has been done before academically in Czech Republic where the used Topic Modeling and Sentiment analysis which is combined into a triangular fuzzy number to be later used for the decision support system component of their system. They utilized this decision support system for municipal decision making. This field was criticized for the disproportionate influence of companies that utilizes smart city solutions where technological innovations are overemphasized and the social dimensions are neglected. This was an area of concern as citizens and local authorities have divergent priorities (Svana et al., 2026).

It utilizes identification of topics from social networks and the sentiment classifications (Positive, Neutral, Negative) expressed toward each topic. Their study stated that it can be used by municipal decision makers to act in the given information in two ways: First to investigate the diversity of sentiments in a single topic for more details from the dynamics of the distribution of sentiment classifications a certain topic had with more diverse sentiment classifications as something worth investigating, and Secondly for resource allocation in

which topics should a municipality commit more time, consultations, and expertise for attending to community needs, more community-aligned information dissemination to the public (Svana et al., 2026).

Synthesis of Literature and Studies, Research Gaps, and Justification

Literature at the international, national, and institutional levels demonstrates the effective use of probabilistic modeling, specifically Latent Dirichlet Allocation (LDA) and contextual transformers, in processing unstructured data into highly interpretable thematic clusters.

Although existing studies have demonstrated significant advancements, critical computational gaps remain. The institutional studies reviewed rely primarily on sensor data, manual user inputs, and GPS hardware, leaving them unequipped to automatically parse high-velocity, multilingual disaster reports. Conversely, international studies successfully integrate Natural Language Processing (NLP) and contextual embeddings, but these methodologies have yet to be rigorously benchmarked for localized Philippine deployment. Local Disaster Risk Reduction and Management Offices (LDRRMOs) currently lack the computational infrastructure to process mixed-syntax Taglish text at scale.

The present study seeks to bridge this gap through a novel comparative methodology. Instead of relying on classical classifiers or manual GPS tracking, the proposed framework introduces a Joint Matrix architecture integrating LDA topic probabilities with XLM-RoBERTa contextual embeddings to preserve interpretability while mapping urgent distress signals. By benchmarking this hybrid system against a state-of-the-art open-weight dense large language model (Qwen3-1.7B-Instruct) operating under few-shot conditions, this research provides the necessary empirical validation to integrate advanced, automated semantic and sentiment modeling of natural disaster-related social media posts.

Research Hypothesis

This study formulates null hypotheses corresponding to the two research questions. The hypotheses address the association between topics and sentiment classifications within each system and the difference in classification correctness between System A and System B relative to stakeholder assessments.

Hypotheses for Research Question 1

Null Hypothesis 1 (H_{01A}): There is no statistically significant association between topics and sentiment classifications in the outputs of System A.

Null Hypothesis 2 (H_{01B}): There is no statistically significant association between topics and sentiment classifications in the outputs of System B.

These hypotheses will be evaluated using the Chi-square test of independence or Fisher's exact test, as appropriate to the categorical data and the assumptions of the statistical tests.

Hypotheses for Research Question 2

Null Hypothesis 3 (H_{02A}): There is no statistically significant difference in the classification correctness of System A and System B relative to stakeholder assessments of the same Facebook posts and comments.

These hypothesis will be evaluated using McNemar's test with continuity correction, based on paired binary correctness outcomes derived by comparing each system's classifications against the corresponding stakeholder assessments.

Definition of Terms

Chi-Square. A non-parametric statistical test used to determine whether there is a significant association between two categorical variables in a sample. It requires a sufficiently large sample size; typically, all expected cell counts should be ≥ 1, and at least 80% of

expected cell counts must be ≥ 5 (Singh, 2020). In this study, it is used to determine whether a significant relationship exists between the topics and sentiment classifications in each Joint Matrix.

Contextual Embeddings. Numerical representations of text generated by language models that capture the contextual meaning of words based on their surrounding text (Barbieri et al., 2022). In this study, contextual embeddings are obtained through XLM-RoBERTa to support the sentiment classification of disaster-related social media text.

Corpus. A collection of textual data gathered for computational analysis (Xiao, 2010). In this study, it refers to the collected disaster-related Facebook posts and comments that serve as the common dataset processed by both System A and System B.

Count Vectorization. A text representation technique that converts textual documents into numerical vectors based on the frequency of words within the documents (Kozhevnikov & Pankratova, 2020). In this study, Count Vectorization is used to represent the collected text data numerically as a valid input for the Latent Dirichlet Allocation (LDA) topic modeling process in System A.

Few-Shot Prompting. An inference paradigm utilized by **System B** (Qwen3-1.7B-Instruct-Q4_K_M), in which the model performs topic modeling and sentiment analysis based on natural-language task descriptions accompanied by task-specific examples (Gadesha, 2024). In this study, it is used as the prompting technique for System B.

Fisher's Exact Test. An exact statistical test used to determine the significance of the association between two categorical variables. Unlike the Chi-Square test, it calculates the exact probability (p-value) of obtaining the observed data configuration directly from the hypergeometric distribution rather than relying on a large-sample approximation (Mays & Stark, 2026). In this study, it is used as an alternative to the Chi-Square test when its assumptions are not satisfied. Since the study uses a $K \times 3$ topic-sentiment contingency

table, the Freeman-Halton extension is utilized for the analysis of larger contingency tables (MetricGate, 2026).

Joint Matrix. A contingency table that presents the joint distribution, frequency counts, and percentages of two categorical variables (NCSS, LLC, n.d.). In this study, it is the computational output of the two systems, represented as a topic-sentiment contingency table of $K \times 3$, where the three columns represent the sentiment classification classes: Positive, Negative, and Neutral, while K represents the number of topics discovered from the corpus by the two systems.

JSON (JavaScript Object Notation). A lightweight, language-independent, text-based data-interchange format designed for structured data representation based on JavaScript object syntax (ECMA International, n.d.). In this study, it is used as a file format for communication with the Large Language Model (LLM) in System B.

Large Language Model (LLM). A type of artificial intelligence model trained on extensive text data to understand and generate human language (Stryker, 2021). In this study, the term refers to the model architecture used in System B, specifically Qwen3, for processing and analyzing disaster-related social media text.

Latent Dirichlet Allocation (LDA). A Bayesian probabilistic generative model used in **System A** to derive underlying topic-related representations and thematic clusters from processed documents (Zang, 2026). In this study, LDA is utilized to discover topics and their corresponding co-occurring words in System A.

McNemar's Test. A non-parametric statistical test used during stakeholder evaluation to analyze paired binary results with continuity correction, assessing whether there is a statistically significant marginal difference between the outputs of two systems (MCP Analytics, 2025). In this study, it is used to identify statistically significant differences in paired binary evaluation outcomes, particularly when the two systems produce different

results based on stakeholder assessments of their topic and sentiment classifications.

Qwen3. An open-weight large language model family developed by Alibaba Group that features dense and mixture-of-experts architectures with built-in hybrid thinking modes (Yang et al., 2025). In this study, it is used as the Large Language Model (LLM) for System B.

Sentiment Analysis / Sentiment Classification. The computational study of opinions, evaluations, and emotions expressed in text (Katrekar, 2019). In this study, each disaster-related post is assigned one of three sentiment labels: Negative (NEG), Neutral (NEU), or Positive (POS). Paired with a post's topic, the label indicates what individuals are discussing and how they feel about it. The three-label scheme is coarse-grained and does not distinguish between specific emotions such as fear and anger.

Taglish. A Philippine code-switching variety characterized by the intra-sentential alternation of Tagalog/Filipino and English lexical and syntactic elements, prevalent in unstructured social media and emergency text feeds (Lesada, 2017). In this study, it refers to the language variety that both systems must process in their respective pipelines.

Topics. Unobserved thematic patterns within a large collection of unstructured text documents (Zang, 2026). In this study, topics refer to clusters of co-occurring words that share a common theme.

UMass Coherence. A topic coherence metric that measures the document-level co-occurrence of a topic's top-N words. Its values are negative, with higher values (closer to zero) indicating more coherent topics (Mimno et al., 2011). In this study, it is used to select K, the number of topics in Latent Dirichlet Allocation (LDA), which subsequently determines the dimensionality of the $K \times 3$ Joint Matrix.

XLM-RoBERTa (Cross-lingual Language Model - Robustly Optimized BERT Pre-training Approach). A transformer-based language model utilized in **System A** to extract

deep contextual language representations from unstructured text, supporting multilingual text processing and analysis (Barbieri et al., 2022).

CHAPTER III

RESEARCH METHODOLOGY

This chapter details the research methodology employed to evaluate and compare the performance of two disaster-related sentiment and semantic analysis models: **System A**, a hybrid LDA + XLM-RoBERTa Joint Matrix System, and **System B**, Qwen3-1.7B-Instruct-Q4_K_M, within the context of the Bulacan and CAMANAVA areas from 2020 to 2025. It establishes a comparative quantitative evaluative research design, detailing the data collection and semi-automated preprocessing of the social media corpus, the architectural frameworks of both systems, and the common measurement metrics used to address the two core research questions. Finally, it outlines the purposive selection of the expert stakeholder panel and the ethical guidelines ensuring participant privacy and data integrity.

Methods of Research Used

Overall Research Design

This study utilizes a comparative quantitative evaluative research design to evaluate the performance of two disaster-related sentiment and semantic analysis systems: **System A**, a hybrid LDA + XLM-RoBERTa Joint Matrix System, and **System B**, Qwen3-1.7B-Instruct-Q4_K_M. Both systems have the same corpus of disaster-related textual data from the Bulacan and CAMANAVA area within the study period. The two systems are designed to be evaluated on multiple complementary measures rather than a single measure of performance. These measures correspond to the two main research questions: topic-sentiment association analysis, and sampled comparative evaluation of their outputs assessed by stakeholders.

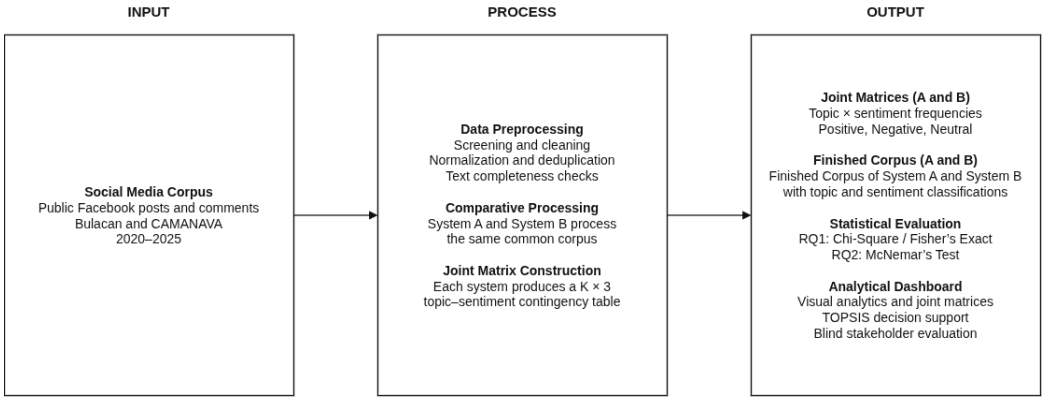


Figure 3.1: Input Process Output(IPO)

This study follows the common comparative pipeline that collects disaster-related textual data, filters, preprocesses, and prepares it for analysis by both systems. **System A** processes this corpus via Count Vectorization, Latent Dirichlet Allocation (LDA), and XLM-RoBERTa-based contextual application, which are then combined into a joint matrix for further evaluation. **System B** processes the same corpus using `Qwen3-1.7B-Instruct-Q4_K_M` through structured prompts for the same sentiment analysis and semantic assessment. The two systems thus fall under a common data source, allowing for a comparative evaluation to proceed.

The study's textual disaster-related data is geographically linked to Bulacan and CAMANAVA, an area consisting of Multiple Cities: Caloocan, Malabon, Navotas, and Valenzuela. During the study period of 2020-2025. The corpus includes social media content from government or disaster-management related public pages relative to the inclusion and exclusion criteria. The methodology identifies Bulacan and CAMANAVA as the geographic scope and specifies Tagalog, English, and Taglish as the principal language context of the textual data.

The study will apply a common measurement approach to the eligible corpus. Each document is processed by both systems under their respective procedures. This allows the resulting sentiments of both to be compared directly.

Research Question 1: Joint Matrix Topic and Sentiment Classification Relationship Analysis

The first question of the study evaluates whether the topic and sentiment classifications from the joint matrix of both **System A** and **System B** demonstrate a measurable association with one another, and how this association describes the dynamics between topics and sentiment classifications and the strength of their relationships analytically. The study implements a chi-square significance test to measure whether the relationship between the topics and sentiment classifications exists, and if the statistical model's assumption does not hold a contingent Fisher's Exact Test is to be employed.

The independent relationship analysis of both systems is conducted with the data inputs and n specifications of potential latent topic topics to create an equally sized joint matrix or contingency table. The analysis shall report relevant relationship measurements of performance: the Chi-Square Statistic and Fisher's Exact Test. The comparison emphasis is on determining analytically the existence of a statistical relationship in their respective joint matrices. Then they are be independently assessed against their acceptability thresholds to conclude whether such a relationship exists and their dependencies with one another for evaluation of which words are being talked about frequently together and how people feel about them, before comparing the two systems again for further insights. Thresholds apply exclusively to independent analysis.

Research Question 2: Sampled Comparative Evaluation

The second question of the study examines both **System A** and **System B** through a common evaluation corpus. The objective is to attain a representative output of the two systems. It is to be structured to ensure representation of outcome patterns where the two systems agree, where they disagree, and cases representing different sentiment levels, where deemed appropriate.

The sampled documents from both systems are blindly evaluated by stakeholders. This

assessment focuses on their own assessment of the same documents and their agreement or disagreement when presented in two blind tests sampled from evaluated outputs that do not specify which is made by **System A** and which is made by **System B**. It is an assessment with human stakeholders. The two systems are evaluated individually in terms of McNemar's test and other appropriate descriptive measures. They are evaluated independently and together when appropriate for further insights. Acceptability thresholds are applied appropriately to non-comparative measures.

Evaluation Framework

The study utilizes a same-measurement comparative evaluation framework in which **System A** and **System B** are applied to the same corpus of disaster-related textual data under comparable evaluation conditions. This framework ensures that differences observed between the two systems are attributable to their respective processing and analytical approaches rather than to the data themselves.

The comparative process begins with the collection and preparation of an eligible disaster-related textual corpus from Bulacan and CAMANAVA. The same processed corpus is then provided to both systems according to their respective procedures. **System A** applies LDA and an XLM-RoBERTa-based joint matrix, while **System B** processes the same textual input via `Qwen3-1.7B-Instruct-Q4_K_M` using a predefined prompting procedure. The resulting sentiment and latent topic outputs are recorded via a common measurement structure established for the study.

The output of both systems is evaluated through two complementary analyses. First, their joint matrices are assessed using performance measures associated with Research Question 1. Second, a randomly selected sample of common outputs is subjected to blind stakeholder evaluation in which their identities are concealed, and their outputs are assessed under the same evaluation conditions.

This framework therefore is a common-input and comparative-analysis sequence. Each

system is evaluated either together or independently relative to mathematical viability and consistency of measures of performance to gain insights and findings. This allows the study to evaluate the independent ability of each system and the relative performance of one over the other with respect to the mathematical assumptions of each evaluation framework.

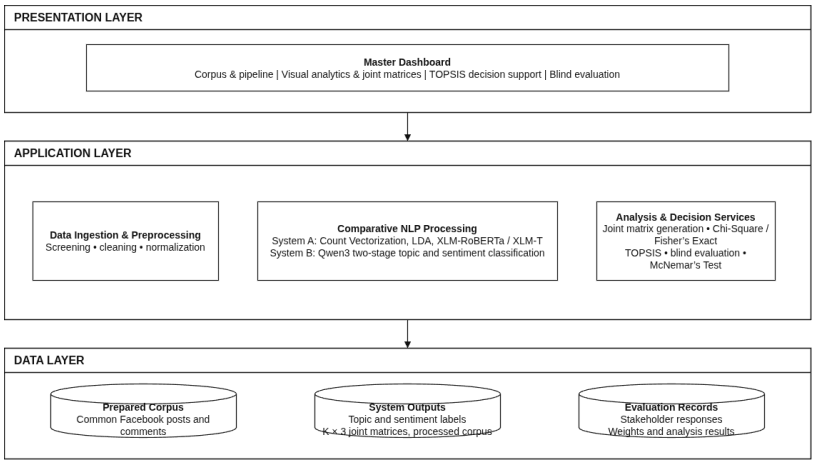


Figure 3.2: Overall Research Pipeline

Data Sources and Corpus

The textual data is collected primarily from publicly accessible Facebook posts and comments related to disasters occurring in Bulacan and CAMANAVA during the study period of 2020-2025. Facebook is the principal data source because posts and comments provide naturally occurring public expressions that may contain information on public sentiment and perceived disaster severity. The corpus includes textual content written in Tagalog, English, and Taglish.

Table 3.1: Target agencies and Facebook source pages used in the comparative flood analysis

Agency / Location	Region	User ID	Facebook URL
Bulacan (General)	Bulacan	bulacan_pdrrmo	https://www.facebook.com/share/1EYVEuEk1i/?mibextid=wwXIfr
City of Malolos	Bulacan	malolos_cdrrmo	https://www.facebook.com/share/1D9yG4Ygq6/?mibextid=wwXIfr
City of Meycauayan	Bulacan	meycauayan_cdrrmo	https://www.facebook.com/share/1Gku63fZAq/?mibextid=wwXIfr
City of San Jose del Monte	Bulacan	csjdm_cdrrmo	https://www.facebook.com/share/1Di3r4DQXf/?mibextid=wwXIfr
City of Baliuag	Bulacan	baliuag_cdrrmo	https://www.facebook.com/share/1rUFil22ejY4/?mibextid=wwXIfr
Calumpit	Bulacan	calumpit_mdrrmo	https://www.facebook.com/share/1jNglrAcf7/?mibextid=wwXIfr
Hagonoy	Bulacan	hagonoy_mdrrmo	https://www.facebook.com/share/1hD7WknPQZ/?mibextid=wwXIfr
Marilao	Bulacan	marilao_mdrrmo	https://www.facebook.com/share/191rS6yyNg/?mibextid=wwXIfr
Bocaue	Bulacan	bocaue_mdrrmo	https://www.facebook.com/share/1BxkjNSc962/?mibextid=wwXIfr
Paombong	Bulacan	paombong_mdrrmo	https://www.facebook.com/share/1C4f8zAHnx/?mibextid=wwXIfr
Obando	Bulacan	obando_mdrrmo	https://www.facebook.com/share/1HmARF9TcY/?mibextid=wwXIfr
Bulakan	Bulacan	bulakan_mdrrmo	https://www.facebook.com/share/1QxQURuK69V/?mibextid=wwXIfr

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Table 3.1: Target agencies and Facebook source pages used in the comparative flood analysis

Agency / Location	Region	User ID	Facebook URL
Bustos	Bulacan	bustos_mdrrmo	https://www.facebook.com/share/1JNzjIK9TH4/?mibextid=wwXIfr
Plaridel	Bulacan	plaridel1_mdrrmo	https://www.facebook.com/share/1FImd5zyCY/?mibextid=wwXIfr
Pulilan	Bulacan	pulilan_mdrrmo	https://www.facebook.com/share/19TQHLEXRX/?mibextid=wwXIfr
Guiguinto	Bulacan	guiguinto_mdrrmo	https://www.facebook.com/share/1djYy9Ghab/?mibextid=wwXIfr
Pandi	Bulacan	pandi_mdrrmo	https://www.facebook.com/share/1GWb8GXpD7h/?mibextid=wwXIfr
Santa Maria	Bulacan	santamaria_mdrrmo	https://www.facebook.com/share/1CF9MHF83a/?mibextid=wwXIfr
San Miguel	Bulacan	sanmiguel_mdrrmo	https://www.facebook.com/share/19fRTlWvuA/?mibextid=wwXIfr
San Ildefonso	Bulacan	sanildefonso_mdrrmo	https://www.facebook.com/share/1c83yUQ9R/?mibextid=wwXIfr
San Rafael	Bulacan	sanrafael_mdrrmo	https://www.facebook.com/share/19VnxKnack/?mibextid=wwXIfr
Angat	Bulacan	angat_mdrrmo	https://www.facebook.com/share/1JbkbF9PpT/?mibextid=wwXIfr
Norzagaray	Bulacan	norzagaray_mdrrmo	https://www.facebook.com/share/1DVq0VAd1B/?mibextid=wwXIfr
Doña Remedios Trinidad	Bulacan	drt_mdrrmo	https://www.facebook.com/share/1G2Yq16Fec/?mibextid=wwXIfr

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Table 3.1: Target agencies and Facebook source pages used in the comparative flood analysis

Agency / Location	Region	User ID	Facebook URL
Valenzuela City	CAMANAVA	valenzuela_cdrrmo	https://www.facebook.com/share/1Gku63fZAq/?mibextid=wwXIfr
Malabon City	CAMANAVA	malabon_cdrrmo	https://www.facebook.com/share/1ELBPLYcVsJ/?mibextid=wwXIfr
Navotas City	CAMANAVA	navotas_cdrrmo	https://www.facebook.com/share/1D6gb4XR6w/?mibextid=wwXIfr
Caloocan City	CAMANAVA	caloocan_cdrrmo	https://www.facebook.com/share/1F78Zh77BW/?mibextid=wwXIfr

Table 3.2: Retained Dataset Schema for data extraction structure

Schema Component	Field Name	Description
Record Identification	`post_id`	Unique identifier assigned to each record.
	`parent_post_id`	Identifier of the original post associated with a comment.
	`row_type`	Identifies whether a record is an original post or a comment.
Temporal Metadata	`timestamp`	Date and time associated with the record.
Geographical Metadata	`region`	Geographic region associated with the record.
	`location`	Specific municipality, city, or barangay associated with the record.
Source Metadata	`source_type`	Identifies the Facebook source category associated with the record.
Textual Content	`text`	Textual content of the original post or comment.
Engagement Metadata	`num_comments`	Number of comments associated with an original post.

Data collection is a semi-automated manual extraction procedure supported by a web scraper

browser extension to aid in extracting the textual data from posts and their comment threads. The relevant Facebook posts and their available comments is identified using the study's own inclusion and exclusion criteria. The web scraper extension assists in the extraction and organization of available data, while manual procedures will identify relevant sources.

Table 3.3: Keywords used for disaster and flood detection in the social media scraper

Category	English Keyword	Tagalog/Local Term	Description
Historical Events	Typhoon Ulysses	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Rolly	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Quinta	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Fabian	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Karding	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Paeng	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Egay	Bagyo	Major tropical cyclone event
Historical Events	Typhoon Carina	Bagyo	Major tropical cyclone event
Water	Flooding	Pagbaha	Flooding condition
Water	Flood	Baha	Flood event
Water	Flash flood	Biglaang baha	Sudden flood surge
Water	Submerged	Lubog sa baha / Lubog	Waterlogged or submerged area
Water	Overflow	Umaapaw / Umapaw	Water exceeds capacity
Water	High Tide	Taib	Coastal water surge
Water	Dam release	Pagpapakawala ng tubig	Controlled drainage from dam
Water	Spillway	Spillway	Floodwater release outlet
Weather	Rainfall	Buhos ng ulan	Heavy rainfall
Weather	Rain	Ulan	Rainfall
Weather	Heavy Rain	Malakas na ulan	Intense rainfall

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Table 3.3: Keywords used for disaster and flood detection in the social media scraper (continued)

Category	English Keyword	Tagalog/Local Term	Description
Weather	Thunderstorm	Kidlat at kulog	Storm with lightning
Weather	Monsoon	Habagat / Amihan	Seasonal wind and rain pattern
Weather	Low Pressure Area	LPA	Weather disturbance
Rescue	Rescue	Saklolo / Sagip	Emergency rescue action
Rescue	Help	Tulong	Assistance request
Rescue	Evacuate	Lumikas	Leaving threatened area
Rescue	Evacuation	Paglikas / Likas	Mass movement to shelter
Rescue	Evacuation Center	Sentro ng paglikas	Shelter for evacuees
Rescue	Trapped	Na-trap / Nakulong	Unable to evacuate
Rescue	Stranded	Naipit	Stuck due to flood or weather
Rescue	Missing	Nawawala	Missing persons report
Rescue	Casualty	Nasawi / Patay	Fatality or death
Rescue	Relief goods	Ayuda	Aid and support delivered
Warning	Advisory	Abiso	Public warning
Warning	Alert	Alerto	Warning alert issued
Warning	Hazard	Panganib	Hazard threat
Warning	Red Warning Level	Red Warning Level	Severe alert status
Warning	Orange Warning Level	Orange Warning Level	Moderate-to-high alert status
Warning	Yellow Warning Level	Yellow Warning Level	Precautionary alert status
Warning	Signal No. 1	Signal No. 1	Weather warning signal
Warning	Signal No. 2	Signal No. 2	Weather warning signal

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Table 3.3: Keywords used for disaster and flood detection in the social media scraper (continued)

Category	English Keyword	Tagalog/Local Term	Description
Warning	Signal No. 3	Signal No. 3	Weather warning signal
Warning	Signal No. 4	Signal No. 4	Weather warning signal
Warning	Signal No. 5	Signal No. 5	Weather warning signal
Threat	Landslide	Pagguho ng lupa	Slope failure event
Threat	Class suspension	Walang pasok / Suspindido ang klase	School closure due to hazard
Threat	Threatening	Nagbabanta	Risk or threat warning
Infrastructure	Angat Dam	Angat Dam	Dam infrastructure
Infrastructure	Bustos Dam	Bustos Dam	Dam infrastructure
Infrastructure	Ipo Dam	Ipo Dam	Dam infrastructure

The data sourcing targets an approximate corpus size of 2,000-5,000 records and above this assumes aggregated entry of posts and its parent comments to a single text for LDA component of system A requiring high character count and quality qualitative data if sample size is small. It sources from publicly available pages of disaster-related agencies around Bulacan and CAMANAVA, with optional inclusions of newer agencies and wider public government pages. The timeframe of 2020-2025 is chosen to obtain relevant qualitative data that is relatively not out of date and is feasible to attain.

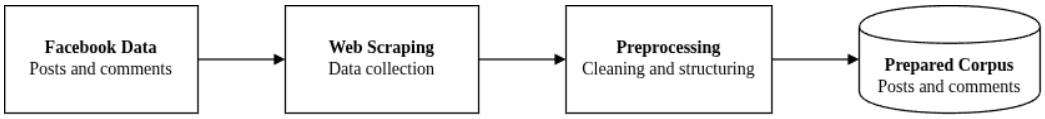


Figure 3.3: Data Collection Pipeline

Inclusion Criteria

Facebook posts or comments are included if they satisfy the following criteria: they are publicly accessible; they are related to a disaster, disaster event, disaster impact, response, recovery, or other

related topics; they are relevant to Bulacan or CAMANAVA; they fall within the 2020–2025 study period; they contain textual content suitable for sentiment and semantic analysis; and they are written in Tagalog, English, or Taglish.

Exclusion Criteria

Facebook posts or comments are excluded when they are not publicly accessible; when they are not related to a disaster, disaster event, disaster impact, response, recovery, or other related topics; when they fall outside the scope of Bulacan or CAMANAVA; when they fall outside the 2020–2025 study period; when they contain no textual content suitable for sentiment and semantic analysis; when they consist entirely of spam, advertisements, promotional materials, and other unrelated content; when they are duplicated records bearing the same content or corrupted text; and when they contain any mentions, hashtags, private information, or emojis.

Table 3.4: Inclusion and Exclusion Criteria for Facebook Posts and Comments

#	Dimension	Inclusion Criteria	Exclusion Criteria
1	Accessibility	Publicly accessible	Not publicly accessible
2	Topical relevance	Related to disaster, disaster impact, response, recovery, or related topics	Unrelated to disasters or related topics
3	Geographic scope	Relevant to Bulacan or CAMANAVA	Outside the scope of Bulacan or CAMANAVA
4	Time frame	Within the 2020–2025 study period	Outside the 2020–2025 study period
5	Text suitability	Text suitable for sentiment and semantic analysis	No suitable textual content
6	Language	Tagalog, English, or Taglish	Other languages
7	Content quality	—	Entirely spam, advertisements, or promotional content
8	Record integrity	—	Duplicated or corrupted records
9	Privacy/formatting	—	Contains mentions, hashtags, private information, or emojis

Together, the resulting eligible records constitute the common corpus used by both **System A** and **System B** under their individual procedures, so that differences in sentiment and topic outputs can be examined under a common data basis.

Data Preprocessing

The collected common corpus undergoes preprocessing before being submitted to **System A** and **System B**, with the intention of removing irrelevant content while preserving linguistic information necessary for sentiment and latent topic analysis. The prepared corpus is sent to both systems to remain consistent with the comparative analysis framework. The records are screened and cleaned based on the inclusion criteria; any content that falls within the exclusion criteria is to be excluded and removed. Text normalization is to be done next to improve consistency; this will remove formatting artifacts, excessive whitespace, and other characters that are not relevant to the study. Record consolidation done afterward to remove remaining duplicates. Finally, the corpus is subjected to textual completeness and analytical suitability checks before being fed to the two systems.

Comparative Systems

This study evaluates two computational systems for disaster-related sentiment and latent topic analysis: **System A**, a hybrid LDA–XLM-RoBERTa Joint Matrix System, and **System B**, Qwen3-1.7B-Instruct-Q4_K_M. Though the two different systems employ different processing approaches, both receive the same prepared disaster-related textual corpus and produce sentiment and latent topic outputs that are evaluated using the common measurement framework of the study. This arrangement allows the study to distinguish differences arising from the analytical approaches of the systems while maintaining a common basis for comparison.

System A: LDA-XLM-RoBERTa Joint Matrix System

System A employs a hybrid approach that combines the topics that Latent Dirichlet Allocation (LDA) found, and the sentiment classifications XLM-RoBERTa produced on the corpus into a joint matrix of topic-sentiment classifications contingency table.

Count Vectorization

Count Vectorization is the process of converting text documents into numerical vectors that allow machine learning models to perform natural language processing (NLP). This starts with removing unnecessary elements such as punctuation, URLs, numbers and text conversions to lowercase. The model processes them by splitting the documents into words so that the model can understand and analyze each word individually, followed by identifying unique words from the preprocessed text that maps text into a numerical vector (Kozhevnikov & Pankratova, 2020).

Vectorization is a function that maps a text document into a numerical vector in a high-dimensional space. This count vectorization is necessary as it is a valid input of LDA.

Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation is a generative probabilistic model of a corpus. The assumption is that each document is represented as a random mixture over topics, and each of these topics is characterized by a distribution of words. Its generative process follows that, for each document d in the corpus:

1. Choose $\theta_d \sim \text{Dirichlet}(\alpha)$.
2. Choose $\phi_k \sim \text{Dirichlet}(\beta)$ for each topic $k \in \{1, \ldots, K\}$.
3. For each of the N_d words $w_{d,n}$:
    a. Choose a topic $z_{d,n} \sim \text{Multinomial}(\theta_d)$.
    b. Choose a word $w_{d,n} \sim p(w_{d,n} \mid z_{d,n}, \beta)$, a multinomial probability conditioned on the topic.

Each document draws its overall topic mixture θ_d from a Dirichlet distribution parameterized by α Given the parameters α and β. The parameters of the LDA get modified depending on their level. Overall, the model has three levels: corpus level, document level, and word level. The parameters α and β are corpus-level parameters and are sampled once in the process of generating a corpus. The variables θ_d are document-level variables and are sampled once per document. Finally, the variables

$z_{d,n}$ and $w_{d,n}$ are word-level variables and are sampled once per word in each document (Blei et al., 2003).

LDA was utilized in System A as a way to discover the topics that people talked about based on the co-occurring words in the given corpus.

Latent Dirichlet Allocation (LDA) K-Topic Selection

The number of K topics is selected empirically through a coherence sweep across $K \in \{2, 3, \ldots, K_{\max}\}$. For each value of K, the model is fitted on the prepared corpus with a fixed random state of 42, batch learning, and 100 maximum iterations, and the mean UMass Coherence score and its respective criterion determined the K-topic quantity of the model.

The UMass coherence score evaluates the quality of the topic models. It relies on the internal corpus of the model, and words are calculated in order of their frequency within the topic; the expected outcome is negative, with values nearest to zero being better. To compute the UMass coherence score:

$$C_{\text{UMass}}(T_k) = \sum_{m=2}^{N} \sum_{n=1}^{m-1} \log \frac{D(w_m, w_n) + 1}{D(w_n)} \tag{3.1}$$

where $T_k = \{w_1, w_2, \ldots, w_N\}$ is the list of N probable words in topic k, $D(w_m, w_n)$ is the number of documents in the corpus containing both w_m and w_n, and $D(w_n)$ is the number of documents containing w_n. This metric relies on co-occurrence statistics gathered from the corpus being modeled and is domain specific (Mimno et al., 2011).

Because the sentiment category has exactly three categories. The selected K—the UMass coherence score nearest to zero among the sampled values—will be the selected amount of K-topics for the LDA configuration and the amount of topics for **System B**'s Stage 1, to preserve the matched dimensionality of their joint matrices required for the comparative evaluation framework.

XLM-RoBERTa (XLM-T)

LDA only gives a platform to conduct topic modeling; it only allows to see which words co-occur relative to a given corpus. It does not tell whether people talk about these sets of words or topics negatively, positively, or neutrally. Sentiment analysis utilizing language models in multilingual

settings makes this possible. XLM-T is a transformer-based model that is an extension of XLM-RoBERTa trained on social media posts on X (formerly Twitter). The model is trained on 198 million social media posts that include English, Tagalog, and many more, with Tagalog/Filipino comprising roughly one million posts (Barbieri et al., 2021).

Following pretraining, XLM-T was fine-tuned for sentiment classification on human-annotated datasets in eight core languages. Tagalog/Filipino is not among these eight fine-tuning languages. Instead, Tagalog sentiment classification is supported via zero-shot cross-lingual transfer (Barbieri et al., 2021). This allows XLM-T to perform the sentiment classifications in a multilingual setup such as this study.

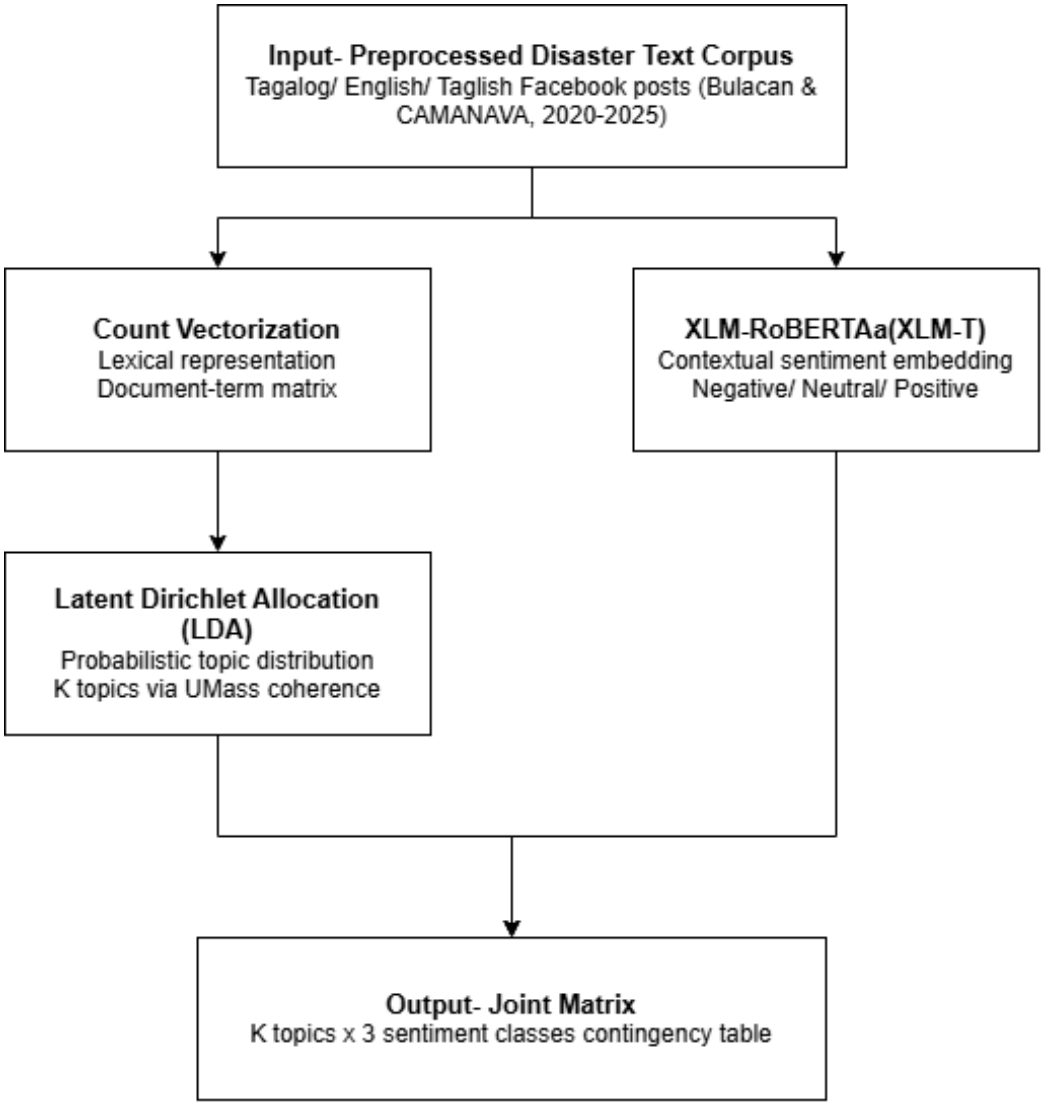


Figure 3.4: System A Architecture

The system generates sentiment classifications using the defined ternary sentiment categories of Negative, Neutral, and Positive, as well as the topics and their respective words. These outputs are fixed before the comparative evaluation.

System B: `Qwen3-1.7B-Instruct-Q4_K_M`

System B employs `Qwen3-1.7B-Instruct-Q4_K_M` as a few-shot language model for disaster-related sentiment and latent topic analysis. Unlike **System A**, which is a hybrid approach utilising multiple representations, **System B** processes the textual inputs directly through a predefined prompting procedure.

Large language models are transformer-based neural networks trained on very large text collections. Their attention mechanism computes each token's representation from the other tokens, so the model uses context from the whole post, which helps with informal and code-switched text(Brown et al., 2020).

System B uses `Qwen3-1.7B-Instruct-Q4_K_M`, a 4-bit quantized, open-weight (Apache 2.0) model. Qwen3 expanded language coverage from 29 to 119 languages and dialects, which is relevant to Tagalog and English but does not by itself show strong Taglish performance. Quantization saves memory at a possible cost in accuracy (Yang et al., 2025). This model was utilized for its multilingual traits and ability to handle the stated constraints of this study.

The prompting procedure specifies the required sentiment categories and topics output format to ensure consistency and that comparative analysis assumptions are respected. For sentiment analysis, the system produces one of the three defined categories: Negative, Neutral, or Positive.

Model Specifications

System B utilizes the Qwen3-1.7B-Instruct model in GGUF format at Q4_K_M quantization. Inference is performed with temperature at 0.1 and a top-p of 1.0. It has a maximum generation length of 2048 tokens for stage 1 and 48 tokens for stage 2. This low temperature reduces sampling variance across documents and supports reproducibility of the classification outputs.

Two-Stage Prompting Architecture

System B operates in two stages: topic discovery and joint classification. This separation is crucial, as the model must first establish the canonical topic taxonomy before it can assign topic labels to individual documents. The number is fixed to the value of the K-topic to match **System A**'s LDA dimensionality, which guarantees that the joint matrix (topic–sentiment contingency table) of both systems has the same matrix shape and is legible for the common measurement framework.

Stage 1: Topic Discovery. In the first stage, a representative sample of 200 documents is drawn from the prepared corpus. This representative sample is presented to the model with instructions to synthesize a canonical taxonomy of distinct K topics. The prompt 1 directs the model to focus on core actions, problems, or topics rather than proper nouns, locations, or geographic identifiers. Each topic must have a short label and a set of keywords extracted from the representative sample.

The model returns a JSON array of topic objects. The prompt enforces strict JSON output with no markdown or explanations. This discovered taxonomy is validated for structural completeness before being used in Stage 2. The final taxonomy is recorded to a CSV file and is used for the common measurement framework.

The discovery sample is drawn using random sampling from the whole prepared corpus. This rule is fixed to prevent bias toward the first entries in the corpus being sampled first.

Stage 2: Joint Sentiment and Topic Classification. In the second stage, each document in the prepared corpus is classified individually. The prompt presents the model with the final taxonomy from Stage 1 and a small set of few-shot examples demonstrating the sentiment decision rule. The model is required to output a single JSON object with two fields: `sentiment` and `topic`.

The sentiment must be exactly one of the following: `POS(Positive)`, `NEG(Negative)`, or `NEU(Neutral)`. The prompt defines the following as:

- `POS`: the text expresses joy, gratitude, hope, relief, or praise;
- `NEG`: the text expresses anger, sadness, fear, frustration, or criticism;

- `NEU`: the text is purely informational, factual, asking for help, or expresses a balanced or mixed view.

The prompt 2 explicitly instructs the model not to assume that factual requests are negative, addressing a common failure mode in disaster text classification. The `topic` field must be exactly one of the label values from the discovered taxonomy. As in Stage 1, the model is instructed to output only valid JSON with no markdown or explanations.

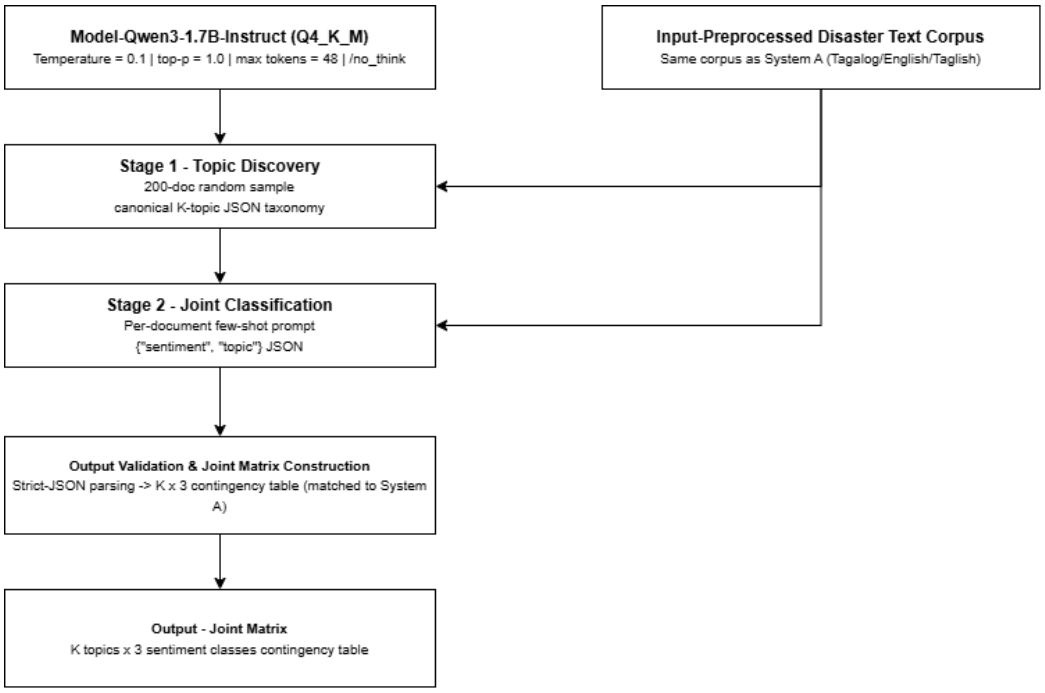


Figure 3.5: System B Architecture

Both systems are therefore applied to the same processed corpus but retain their individual internal procedures. Their outputs are to be subjected to the same evaluation framework for independent assessment and comparative analysis.

End Products of System A and System B: The Joint Matrix

In order for both systems to undergo the same measurement comparative analysis, both their latent topic matrices and their sentiment classification matrices are fused together onto a single joint matrix that has the same dimension and size via contingency table application. This will allow

quantifiable measurements of performance to be performed on both systems' outputs independently and relative to one another.

Contingency tables allow the evaluation of the joint distributions, frequency counts, and percentages of two categorical variables. Formally, the table variables define the rows and columns, one variable defining the rows of the table and another variable defining the columns of the table (NCSS, LLC, n.d.).

Let $T = \{\tau_1, \ldots, \tau_M\}$ be the row variable of topics with M topics, and $S = \{\sigma_1, \ldots, \sigma_N\}$ be the column variable of sentiment classifications with N classes. Then an $M \times N$ contingency table with marginal totals is defined as

$$J = \begin{pmatrix} j_{11} & j_{12} & \cdots & j_{1n} & n_{1n} \\ j_{21} & j_{22} & \cdots & j_{2n} & n_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ j_{m1} & j_{m2} & \cdots & j_{mn} & n_{mn} \\ n_{m1} & n_{m2} & \cdots & n_{mn} & n \end{pmatrix} \tag{3.2}$$

where each cell entry is defined as

$$j_{mn} = \left| \{ d \in \mathcal{D} : t(d) = \tau_m \ \wedge \ s(d) = \sigma_n \} \right| \tag{3.3}$$

with $t(d)$ representing the assigned latent topic of document d and $s(d)$ representing its assigned sentiment classification.

Measurement and Evaluation

The information collected from the two systems is processed, coded, and analysed with suitable performance measures that correspond to the two research questions of this investigation. System A, the Hybrid LDA + XLM-RoBERTa Joint Matrix System, and System B, $Qwen3 - 1.7B - Instruct - Q4_K_M$, is assessed using the exact evaluation framework to ensure comparative integrity in the evaluation results. Each statistic runs separately on each system prior to comparing and making

relative findings.

System Implementation: The Analytical Dashboard

Guided by sensemaking theory, the backend computational pipelines are encapsulated within a localized analytical web dashboard. While System A and System B perform the heavy mathematical workload of vectorization and probabalistic modeling. The dashboard provides the necessary visual interface for human-in-the-loop validation. It translates abstract joint matrices and evaluated corpus into interactive charts and blinded evaluation forms, enabling local stakeholders to visually verify the alignment between disaster topics and community sentiments.

The dashboard will have these primary modules:

1. **Corpus and Pipeline Ingestion Module.** This module handles the loading of the pre-processed, cleaned corpus and executes the backend pipelines of System A and System B. It manages the generation of the Joint Matrices and stores the resulting sentiment and topic classifications in structured data formats (CSV) for frontend rendering.

2. **Visual Analytics and Joint Matrix Module.** This module renders the mathematical outputs of the study. It displays interactive bar charts for sentiment distributions, topic-word probability heatmaps, and the $M \times N$ Joint Matrix contingency tables. This allows the researchers and technical experts (Cohort B) to visually validate the coherence of the LDA topics and the alignment of the sentiment classifications without parsing raw code.

3. **Decision Support System Module.** Uses TOPSIS to evaluate and rank topics based on sentiment distributions and stakeholder-defined weighting, helping local stakeholders identify which topics warrant further investigation and prioritization of resources.

4. **Blind Evaluation Instrument Module.** To execute Research Question 2, the dashboard features a specialized evaluation interface. It randomly samples documents from the corpus and presents the system-generated sentiment and topic classifications side-by-side in a blinded format (labeled only as “Output X” and “Output Y”). The stakeholder panel (Cohort A and B) interacts directly with this web interface, utilizing a assessment format GUI to record their

agreement or disagreement. The dashboard automatically logs these interactions into a secure database for the subsequent McNemar's Test and other computations.

By utilizing a lightweight python web framework, the dashboard ensures the the system remains entirely dependent on local backend environment satisfying the need for an interactive visual analytical system.

Decision Support System (DSS)

Decision Support System (DSS) was first introduced as a system designed to aid decision makers in semi-structured issues that algorithms cannot resolve. It served to increase the managers capabilities rather than replace them in the decision making process. DSS provides a quantitative basis that supports managerial decisions instead of solely relying on personal intuition and experience (Rashidi et al., 2018).

Technique for Order Performance by Similarity to Ideal Solution (TOPSIS) is one type of DSS. It attempts to calculate alternative values by calculating the positive ideal value which is far from the negative ideal value. The positive ideal value is obtained from maximizing the value of the score and minimizing the value of the cost.A negative criteria value maximizes the cost but minimizes the score. It can be said that the positive ideal solution is all the best score of the criteria, while the negative ideal solution is the bad score of the criteria (Daniati & Utama, 2021). In the context of this study the topics will be the possible alternatives while the sentiment classifications will be the criteria relative to the specifications and weighing emphasis.Steps into conducting a TOPSIS decision support system is as follows:

1. building a performance decision matrix and its subsequent normalized values. This was already done in the joint matrix part of the pipelines for both systems.

2. applying the weighing to the normalized decision matrix. The normalized versions of the joint matrices of the two systems will be multiplied to the weighing class of 3 values relative to the end user specification on which sentiment class they want to emphasis in their rankings.

3. attain the positive and negative ideal solutions of the matrix as follows:

$$A_k^+ = (\max_m j_{mk} | k \in N), (\min_m j_{mk} | k \in N'), (k = 1, 2, \ldots, n)$$

$$A_k^- = (\min_m j_{mk} | k \in N), (\max_m j_{mk} | k \in N'), (k = 1, 2, \ldots, n)$$

this will give us the best and worst ideal solutions. This is relative to the weighing of the criteria (sentiment classifications).

4. Calculate the distance from each alternative decisions to the positive ideal solution and the negative ideal solution using n dimensional Euclidean distance.

$$D_m^+ = \sqrt{\sum_{k=1}^{n} (j_{mk} - A_k^+)^2}$$

$$D_m^- = \sqrt{\sum_{k=1}^{n} (j_{mk} - A_k^-)^2}$$

This will give us the distance of all alternative decisions relative to the positive ideal solution and negative ideal solution

5. Calculates the relative proximity to the ideal solution as follows:

$$R_m = \frac{D_m^-}{D_m^+ + D_m^-}, (m = 1, 2, .., \text{total rows})$$

After calculating the alternative proximity and the value of R_m, the next step is to rank the alternatives and sort the values of R_m. The alternative with the highest score is the top choice (topic) given the criteria (sentiment classifications) that is to be emphasized. a value closer to 1 implies the alternative choice is nearer to the positive ideal solution and farther from the negative ideal solution. A value closer to 0 is the opposite (Daniati & Utama, 2021).

This is utilized as the DSS component of the front-end system as it is used to potentially help local stakeholders find topics that has to be investigated due to frequency distribution of their sentiment classifications (high levels of negative and positive sentiment but minimal neutral sentiments, high concentration of sentiment classifications sentiments towards a single class and such) the TOPSIS

methodology allows this to be done systematically with a unified algorithm regardless of weighing of the criteria.

It also helps local stakeholders make informed decisions on which topics should they focus their time, consultations, expertise and other resources to have an aligned response with what people talk about in social networks and how they feel talking about it.

Waterfall Development Methodology

The Waterfall model is a plan-driven software process in which development moves through requirements, design, implementation, verification, and maintenance in sequence (Sommerville, 2016). The model suits projects whose requirements are well understood and unlikely to change, and it produces clear documentation at the end of each phase (Pressman & Maxim, 2020).

SAKLOLO fits these conditions. The disaster types, sentiment categories, the two systems, the evaluation metrics, and the inclusion and exclusion criteria are defined in this proposal before implementation begins. Because the study compares two fixed systems on a fixed corpus, changes in the middle of development would also weaken the comparison.

The phases map onto the study as follows.

Requirements analysis. The corpus criteria (public Facebook posts and comments in Tagalog, English, and Taglish about natural disasters in Bulacan and CAMANAVA, 2020 to 2025), the common output format, and the evaluation metrics are fixed. The deliverable is the finalized scope and evaluation plan.

System design. The System A and System B architectures, the Joint Matrix, and the dashboard layout are specified. The deliverable is the architecture diagrams and the data schema.

Implementation. The data pipeline (collection, preprocessing), the two systems, the evaluation scripts, and the dashboard are built. The deliverable is the working prototype.

Testing and verification. Outputs are checked against the common format, both systems are run on the same corpus, the statistical analyses are computed, and the blind stakeholder evaluation

is conducted. The deliverable is the results tables and test report.

Maintenance. Documentation is completed and recommendations for future work are recorded. Because the study excludes real-time integration into DRRM command centers, maintenance is limited to documentation and handover.

A known weakness of Waterfall is its limited flexibility once a phase is complete (Sommerville, 2016), which is reduced here by fixing the requirements and metrics in advance.

System Flowchart

Research Question 1: Joint Matrix Latent Topic Structure and Sentiment Classification Relationship Analysis

Regarding Research Question 1, an assessment is conducted to determine whether a relationship exists between the respective joint matrices of **System A** and **System B**. This must be done to analytically justify that the two systems have the ability to capture the topics and sentiments of Facebook posts during natural disasters, and their quantifiable measures of performance in doing so.

Bivariate Analysis. As a statistical method, bivariate analysis serves as an analytical tool that determines the empirical relationship between two variables, their interaction with one another, trends, and possible causal relationships. It aims at examining the relationship between two variables to gain insight into whether such a relationship exists. By examining this, it allows a quantifiable understanding of two variables (Ben, 2024). Since our joint matrix is composed of topics and sentiment classifications, this can be interpreted as two variables that the study seeks to analytically determine whether their joint matrix is significantly associated.

Within this investigation, the system-generated topics and sentiment classifications from the joint matrices of both systems serves as the two variables evaluated for bivariate analysis. Prior to performing further bivariate analysis using the Chi-Square Statistic and Fisher's Exact Test, the output is in the form of a cross-tabulation to create a structured framework of computation in matrix format. The joint matrix itself is a form of bivariate analysis, as it is a cross-tabulation or contingency

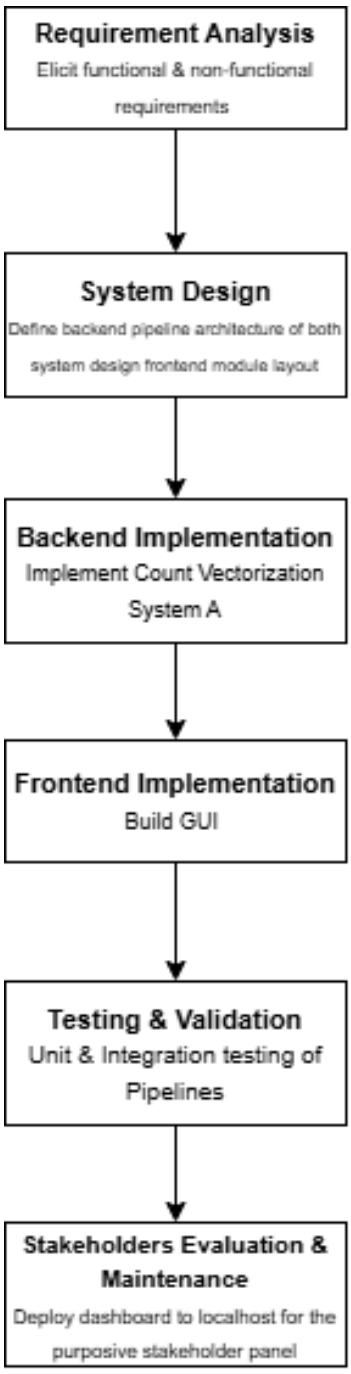


Figure 3.6: Waterfall Development Model

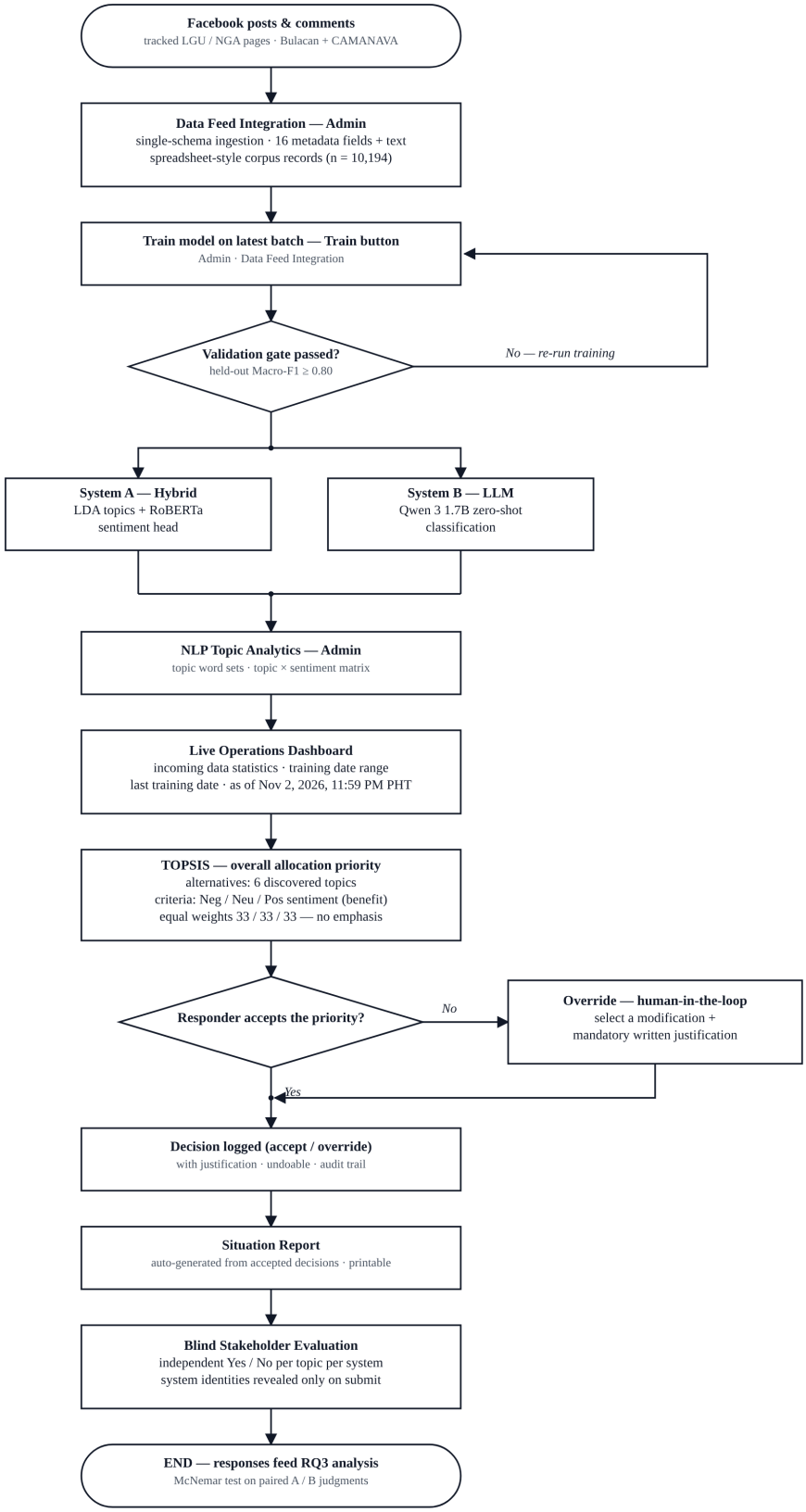


Figure 3.7: System Overview

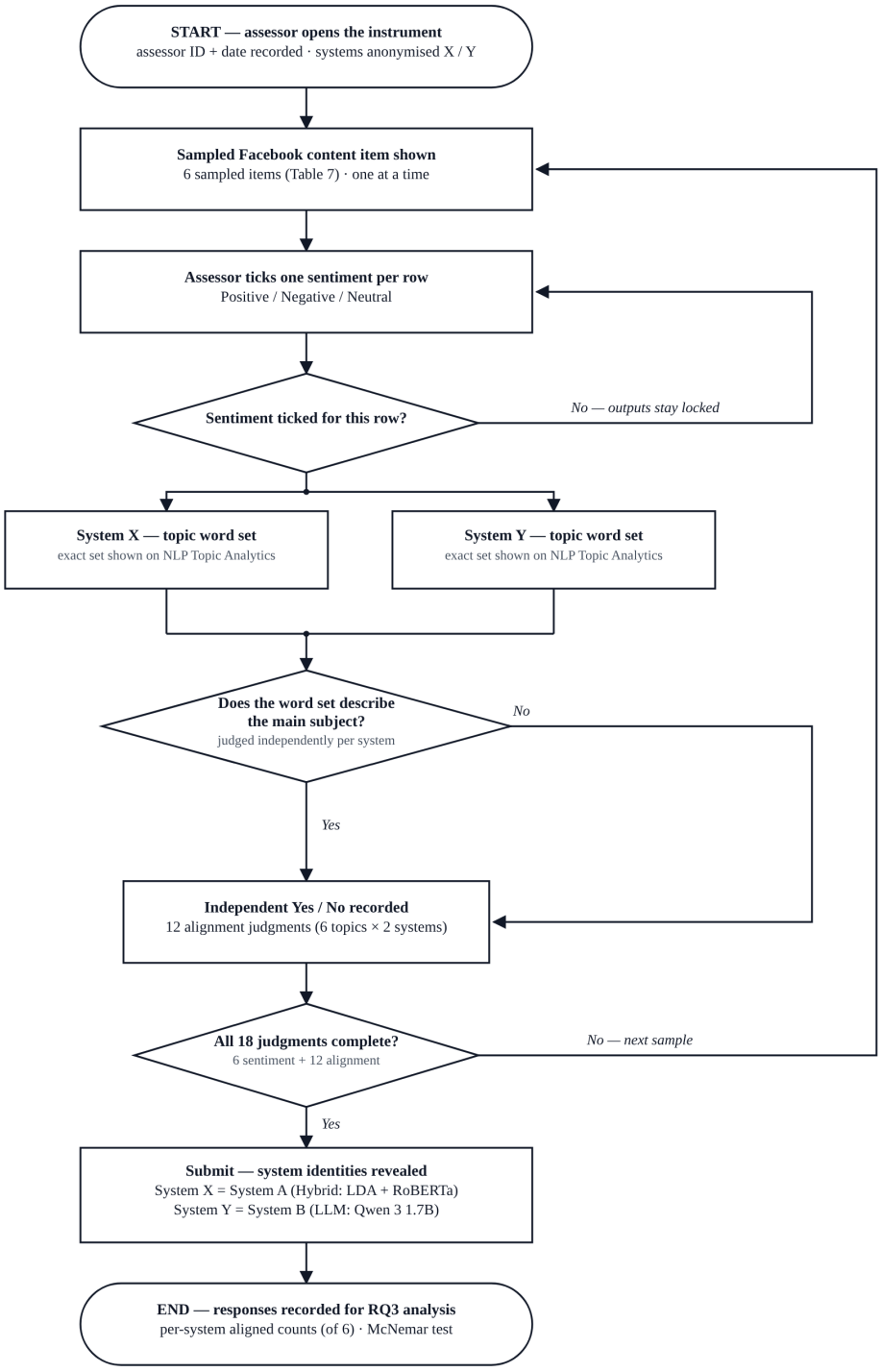


Figure 3.8: blind evaluation

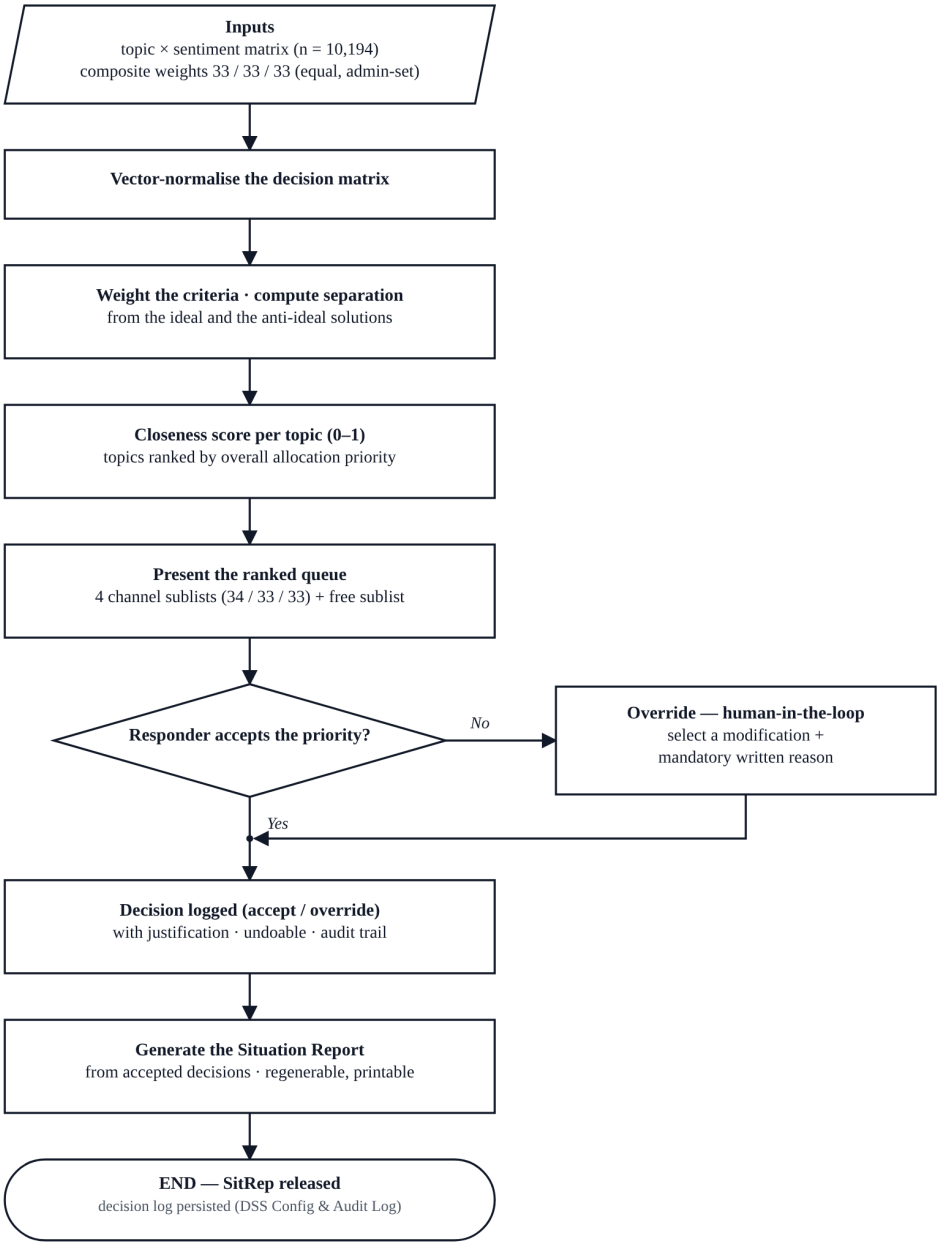


Figure 3.9: TOPSIS layer

table that allows to see the frequency distribution of both variables in the same format and gives insight into their associations (Ben, 2024).

Chi-Square Statistic Test. The chi-square test is used to determine whether there is a significant difference between observed frequencies and expected frequencies and, in contingency tables, the relationship between categorical variables. It is used to determine whether a significant relationship exists between two or more variables. This association can be negative, positive, or an absence of association. To test whether an association exists, the null hypothesis is set such that no association exists and the alternative hypothesis that an association exists. To decide whether to reject or fail to reject the null hypothesis, first compute the chi-square statistic (χ^2) and obtain the critical value to determine whether to conclude that no association exists or that an association exists.The formula to attain χ^2 is

$$\chi^2 = \sum_{i=1}^{R} \sum_{j=1}^{C} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \tag{3.4}$$

where R is the number of rows of the joint matrix (the topics), C is the number of columns of the joint matrix (the sentiment classifications), O_{ij} are the observed frequencies in the joint matrix—the actual raw count of Facebook posts that either of the two systems assigns to latent topic i and sentiment classification j—and E_{ij} are the expected frequencies, or the calculated marginal totals of the matrix if the null hypothesis is true. To attain E_{ij}, the formula is

$$E_{ij} = \frac{R_i\, C_j}{n} \tag{3.5}$$

For determining thresholds, χ^2 is tested against the chi-square distribution table; to attain that, we first need the degrees of freedom df:

$$df = (R - 1)(C - 1) \tag{3.6}$$

After this, a Python function is used to determine the critical value; it takes the contingency table as its input, and if the associated p-value is less than the chosen significance level, the null hypothesis is rejected and the joint matrix is deemed to have a significant association (Singh, 2020).

Fisher's Exact Test. The Chi-Square tests needs to be applied with caution for contingency tables that might have cells that has low frequencies as the statistical model itself has a continous distribution while the underlying data might be discontinuous. It is stated that for Chi-Square test to be applied several assumptions are to be followed:

1. all cells of the contingency table must be > 1
2. atleast 80% of all cells should be > 5

This limitations by the Chi-Square is addressed by the Fisher's Exact Test. It utilizes the hyper-geometric distribution to calculate the probabilities. When utilizing this model it is conservative in achieving a fixed significance level due to the hyper-geometric distribution being highly discrete in small samples. The model can be extended beyond the 2x2 contingency table by utilizing the Freeman-Halton extension(Mays & Stark, 2026).

$$P = \frac{\prod_{i=1}^{R} R_i! \prod_{j=1}^{C} C_j!}{n! \prod_{i=1}^{R} \prod_{j=1}^{C} x_{ij}!} \tag{3.7}$$

Conditioned on fixed row totals R_i and columns total C_j. It also has N as grand total of the table and n_{ij} for the count in cell (i, j) (MetricGate, 2026).

Acceptability Thresholds

The resulting performance measure is examined to confirm whether a relationship exists on the topics and sentiment classifications within the joint matrix of both **System A** and **System B**.

A $p < 0.05$ conventional rejects the null hypothesis for both Chi-Square and Fisher's Exact Test implying there is a relationship between variables and population (Mays & Stark, 2026). This performance measures is applied independently to both **System A** and **System B** before a relative comparative analysis that evaluates their relative performance with one another.

Research Question 2: Blind Stakeholder Evaluation

Research Question 2 evaluates the stakeholder assessment regarding the outputs generated by **System A** and **System B**. From the common evaluation corpus, a randomly selected sample is

examined through a blind evaluation procedure in which the order of the systems' outcomes is randomized. A simple majority rule determines the aggregation rule for each document, because the identity of the system making each output is not revealed to the evaluators. When this evaluation takes place, agreement, differences in ratings, and paired assessment outcomes are considered. Each evaluator is expected to evaluate 30–50 randomly sampled documents drawn from the outputs of **System A** and **System B**the binary outcomes on the sentiment is to match if the stakeholder's sentiment classification is equal to the respective sentiment classifications of a system to that same item and for topics it is direct answer from the assessment.

McNemar's Test. For stakeholder evaluation, paired binary results are examined using McNemar's Test, which is suitable for systematic comparisons when identical assessors evaluate two systems in two categories (MCP Analytics, 2025). The test statistic incorporates continuity correction:

$$\chi^2_{\text{McNemar}} = \frac{(|b - c| - 1)^2}{b + c} \tag{3.8}$$

In the calculation, b indicates cases where **System A** succeeds while **System B** fails, and c indicates cases where **System B** succeeds while **System A** fails. This comparison aims to reveal operational differences, with the p-value being analyzed to assess the statistical significance of the differences between the outcomes of the two systems.

Acceptability Thresholds

The resulting performance measures are examined to confirm whether a statistically significant difference exists between the paired outcomes of **System A** and **System B** when assessing overall performance.

A McNemar's Test at the 0.05 significance level is the minimum acceptability threshold (MCP Analytics, 2025); that is, a system has alignment of the sentiment classification and topic identification of stakeholders around natural disasters in Bulacan and the CAMANAVA area in the same sampled facebook posts or comments in the processed corpus, as it indicates a detection of possible differences between the system and the assessments stakeholders around the same sampled items. These performance measures are applied to both **System A** and **System B**.

Respondents and Expert Evaluators of the Study

The development and comparative evaluation of the SAKLOLO. natural language processing pipeline requires highly specialized domain expertise rather than general population sampling. Consequently, the study utilizes a purposive sampling design to form a Blind Evaluation Stakeholder Panel.

Exclusion of Population-Based Sample Size Formulas

When determining the number of evaluators for natural language processing (NLP) architectures, traditional probabilistic calculations (such as Cochran's or Slovin's formulas) are mathematically and methodologically incompatible with the research design. These formulas are designed for epidemiological or sociological surveys where the goal is to achieve a representative demographic cross-section of a generalized population. They assume that every member of the population has an equal probability and capacity to provide valid data.

The evaluation of the proposed framework, however, is not a demographic survey but rather an experimental computational benchmark. The study requires human evaluators to audit the classification outputs of a hybrid pipeline (LDA + XLM-RoBERTa) and an open-weight transformer $Qwen3-1.7B-Instruct-Q4_K_M$.

Evaluating contextual embeddings, coherence, and algorithmic classification disparities requires calibrated judgment from domain specialists. Using a purposively selected expert panel provides the controlled conditions necessary to execute specific non-parametric statistical validations (such as McNemar's test and rank correlation coefficients) to detect genuine differences in model configurations without introducing data dilution.

Blind Evaluation Stakeholder Panel (System Evaluation)

To execute the blind comparative evaluation between the lightweight hybrid model (**System A**: LDA + XLM-RoBERTa) and the state-of-the-art open-weight (**System B**: $Qwen3-1.7B-Instruct-Q4_K_M$), a total panel of 10 to 20 evaluators will be recruited. This panel is divided equally into two specialized cohorts.

Table 3.5: Cohort A and Cohort B Descriptions

Cohort A: DRRM Operational Officers (n = 5–10)	**Cohort B: Academic and Technical Experts** (n = 5–10)
Source. Active personnel, incident commanders, and communication dispatchers from the Provincial and Municipal Disaster Risk Reduction and Management Offices (DRRMOs) of the Bulacan and CAMANAVA corridors.	*Source.* Faculty members and researchers specializing in Applied Mathematics, Computer Science, and Data Analytics, specifically targeting the Mathematics Department of the College of Science and the allied faculty of the CICT at Bulacan State University.
Operational Role. These practitioners evaluate the practical applicability, operational fidelity, and functional alignment derived from the models without knowing which system generated the output. They provide judgment on whether the extracted sentiment and disaster urgency accurately reflect actual field conditions.	*Operational Role.* These experts assess the mathematical validity and thematic coherence of the extracted topic probabilities and contextual embeddings, ensuring the algorithms operate within accepted standards of information theory and machine learning rigor.

Table 3.6: Summary of Study Participants

Participant Group	**Target Size (n)**	**Sampling Method**	**Target Affiliation**	**Research Function**
DRRM Practitioner Panel	5–10	Purposive	Bulacan and CAMANAVA DRRM Offices	Blind evaluation of system outputs; assesses operational applicability and relevance.
Academic Technical Panel	5–10	Purposive	Mathematics and Computer Science Faculty (College of Science / CICT)	Technical validation of thematic coherence, joint embeddings, and comparative algorithmic quality.
Total Panel	**10–20**	**Purposive**	—	—

In software engineering and machine learning benchmark literature, evaluating models against human baselines relies on expert panel auditing rather than large-scale probability sampling. A focused panel of calibrated domain specialists provides the necessary statistical power to execute paired nonparametric tests while eliminating evaluation fatigue and ensuring reliable judgments.

Research Instrument

The research instrument of the study is a structured blind evaluation form. It assess the agreement of stakeholder assessments with the outputs of **System A** and **System B**. The instrument as shown in table 7 exhibits the sampled Facebook posts or comment from the processed corpus from both systems.

For sentiment assessment, stakeholders classifies first the sentiment of each sampled content as Positive, Neutral, or Negative. These assessments serve as the human reference for determining the alignment of the sentiment classifications produced by both systems.

For topic assessment, the instrument presents the topic representations generated by both systems through their respective sets of co-occurring words, labeled as System X: Topic Words and System Y: Topic Words. Stakeholders assess whether each topic representation adequately describes the main subject of the sampled content by selecting either Yes or No.

The evaluation form records the stakeholder's assessment, the topic-alignment judgments for both systems, and their corresponding sample and system identifiers. These responses are matched with the respective system outputs and converted into agreement observations for subsequent statistical analysis. The same instructions, response categories, and evaluation conditions will be applied to all participating stakeholders to maintain consistency across observations. While System Identifiers remains on the researcher's for tracking the random allocations in System X: Topic Words and System Y: Topic Words utilized to conceal identity and assumptions a single column represents a output of a single system.

Ethical Considerations

This study observes ethical considerations for the safety of participants and maintains the integrity of this paper throughout the development of *SAKLOLO: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA*. The researchers follow ethical guidelines such as collecting data from identified sources, obtaining consent, ensuring the confidentiality of participants and voluntary participation, using social media responsibly, and maintaining integrity in presenting the research results for transparency. Appropriate procedures are established to maintain confidentiality and to protect the privacy of all participants in this study. The study adheres to the following ethical principles:

Data Privacy and Confidentiality.

Republic Act No. 10173, or the Data Privacy Act of 2012, establishes the rights of individuals needed to protect their personal information, and the organization is responsible for the personal information of participants (Republic of the Philippines, 2012). The needed disaster-related report data is processed only if needed, solely for this study. The study excludes unnecessary personal information. The study focuses on collecting disaster-related information from identified sources such as social media and DRRM reports in Bulacan and CAMANAVA. This data is filtered to remove irrelevant data and personal information for the privacy and accuracy of the study. All collected data are for academic research and model validation purposes only (National Privacy Commission, 2016).

Voluntary Participation and Informed Consent.

Participants are informed about the purpose and procedure of this study prior to their involvement. Participants in this study are voluntary; they have the right to decline to participate in this study without penalty. To ensure accuracy and prevent biases during evaluation, a protocol will be established wherein evaluators are anonymized without being informed whether the predictions were from the LLM baseline model or the hybrid pipeline system (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979).

Confidentiality of Participants.

The identity of participants and personal information of stakeholders are kept confidential in this study. The evaluation process uses a blind evaluation procedure so that the system does not reveal the identity of evaluators. Possible data that can identify participants are excluded from the results to prevent biases and to protect the participants. In accordance with the Data Privacy Act of 2012 (National Privacy Commission, 2016; Republic of the Philippines, 2012), the identity of participants are protected against alteration and unauthorized access.

Responsible Use of Social Media Data.

Only data relevant to the study is collected, such as disaster-related data from publicly accessible social media of DRRM offices. The usernames of commenters from social media and other personal identifiers are excluded from the data collection process. The researchers ensure that social media will be used responsibly and solely for academic purposes, not to harm anyone. Moreover, all collected data from social media is anonymized and processed in accordance with ethical research standards (National Privacy Commission, 2016).

Research Integrity.

The researchers of this study uphold transparency, honesty, and objectivity throughout the development of the study. All data related to this study, such as evaluations and results, is properly collected and studied accurately without falsification and fabrication. This study is developed ethically and honestly, and results are presented with integrity whether the outcome is expected or not (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979). This study ensures reliability, credibility, and transparency. Moreover, citations to all sources such as DRRM reports, social media posts, and other materials is properly acknowledged within the paper.

REFERENCES

Adriano, M. N. (2023, January). *Contextualization of grade 3 instructional materials: Basis for a design of bulacan tagalog-based science materials* [Doctoral dissertation]. https://doi.org/10.5281/zenodo.7317449

Anggara, Y. Y., Sumunar, D. R. S., Khotimah, N., & Hidayatulloh, S. (2025). Analyzing public sentiment on the 2024 'galodo' disaster using natural language processing (nlp). *International Journal of Geography, Social, and Multicultural Education*, *3*(2), 1–12. https://journal.unesa.ac.id/index.php/ijgsme/article/download/39883/14342

Balkir, E., Sadrzadeh, M., & Coecke, B. (2016). Distributional sentence entailment using density matrices. In *Lecture notes in computer science* (pp. 1–22). Springer. https://doi.org/10.1007/978-3-319-28678-5_1

Barbieri, F., Anke, L. E., & Camacho-Collados, J. (2021). Xlm-t: Multilingual language models in twitter for sentiment analysis and beyond. *arXiv preprint arXiv:2104.12250*. https://arxiv.org/abs/2104.12250

Barbieri, F., Espinosa Anke, L., & Camacho-Collados, J. (2022). XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond. *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 258–266. https://aclanthology.org/2022.lrec-1.27

Ben, A. (2024). Bivariate analysis techniques: Correlation, regression and their applications. *Mathematica Eterna*, *14*(2), 224. https://doi.org/10.35248/1314-3344.24.14.224

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, *3*, 993–1022.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, *33*, 1877–1901.

Choirul Rahmadan, M., Nizar Hidayanto, A., Swadani Ekasari, D., Purwandari, B., & Theresiawati. (2020). Sentiment analysis and topic modelling using the lda method related to the flood disaster in jakarta on twitter. *2020 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS)*, 126–130. https://doi.org/10.1109/icimcis51567.2020.9354320

Daniati, E., & Utama, H. (2021). TOPSIS in decision-making framework based on Twitter sentiment analysis. *2021 4th International Conference on Information and Communications Technology (ICOIACT)*, 268–273. https://repository.unpkediri.ac.id/4755/1/erna-1570746548%20stamped-e.pdf

De Guzman, J. L. D., Dela Cruz, J. L., Diaz, K. R. C., & Santos, R. D. (2019). *Readi: A modern way of saving lives using mobile application: Real-time emergency alerts and disaster information* [Master's thesis, Bulacan State University] [Unpublished undergraduate thesis].

ECMA International. (n.d.). *Introducing JSON* [Based on the ECMA-404 JSON Data Interchange Standard]. Retrieved October 31, 2026, from https://www.json.org/json-en.html

Gadesha, V. (2024, August). *What is few shot prompting?* IBM. Retrieved October 31, 2026, from https://www.ibm.com/think/topics/few-shot-prompting

Gorro, K. D., Baguia, G. A., & Ali, M. F. (2021). An analysis of disaster risk suggestions using latent dirichlet allocation and hierarchical dirichlet process (nonparametric lda). *2021 The 9th International Conference on Information Technology: IoT and Smart City*, 1–7. https://doi.org/10.1145/3512576.3512608

Inoferio, M. A. V., Hernandez, H. J. A., & Santos, R. A. P. (2025). The role of social media in philippine disaster management: A systematic literature review. *2025 Sustainability,*

Environment, and Energy (SEE) Research Conference. https://animorepository.dlsu.edu.ph/conf_shsrescon/2025/paper_see/33/

Katrekar, A. (2019, December). *An introduction to sentiment analysis* (White paper). GlobalLogic. https://globallogic.com

Kozhevnikov, V., & Pankratova, E. (2020). Research of the text data vectorization and classification algorithms of machine learning. *Theoretical & Applied Science*, *85*, 574–585. https://doi.org/10.15863/TAS.2020.05.85.106

Lesada, J. D. (2017). *Taglish in Metro Manila: An analysis of Tagalog-English code-switching* [Bachelor of Arts with Honors thesis]. University of Michigan. https://backend.production.deepblue-documents.lib.umich.edu/server/api/core/bitstreams/142edfff-85ab-44a4-86f4-e14c9b9f72b8/content

Manalo, J. J. A., Matic, J. I. G., Osdaña, R. C. L., & Salalac, C. D. R. (2024). *Lokamatika: Two-way location sharing and automated emergency response systems with trilateration and kinematics* [Master's thesis, Bulacan State University] [Unpublished undergraduate thesis].

Mays, S., & Stark, S. (2026). The use of Fisher's exact test in contingency table analysis in palaeopathology [Epub 2026 Feb 6]. *International Journal of Paleopathology*, *52*, 135–139. https://doi.org/10.1016/j.ijpp.2026.01.005

MCP Analytics. (2025). Mcnemar's test for paired before-and-after data. https://mcpanalytics.ai/articles/mcnemars-test-practical-guide-for-data-driven-decisions

MetricGate. (2026, June). *Freeman-Halton exact test (r x c) calculator*. MetricGate. Retrieved October 31, 2026, from https://metricgate.com/docs/freeman-halton-exact-test/

Mihunov, V. V., Jafari, N. H., Wang, K., Lam, N. S. N., & Govender, D. (2022). Disaster impacts surveillance from social media with topic modeling and feature extraction: Case of hurricane harvey. *International Journal of Disaster Risk Science*, *13*(5), 729–742. https://doi.org/10.1007/s13753-022-00442-1

Mimno, D., Wallach, H., Talley, E., Leenders, M., & McCallum, A. (2011). Optimizing semantic coherence in topic models. *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, 262–272.

Montenegro, C., Ligutom, C., Orio, J. V., & Ramacho, D. A. M. (2018). Using latent dirichlet allocation for topic modeling and document clustering of dumaguete city twitter dataset. *Proceedings of the 2018 International Conference on Computing and Data Engineering*, 1–5. https://doi.org/10.1145/3219788.3219799

National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. (1979). *The belmont report: Ethical principles and guidelines for the protection of human subjects of research* (tech. rep.). U.S. Department of Health, Education, and Welfare. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html

National Privacy Commission. (2016). Advisory on the data privacy act of 2012. https://privacy.gov.ph/

NCSS, LLC. (n.d.). Contingency tables (crosstabs / chi-square test). https://www.ncss.com/wp-content/themes/ncss/pdf/Procedures/NCSS/Contingency_Tables-Crosstabs-Chi-Square_Test.pdf

Pagudpud, M. V. (2020). Public sentiments in cagayan valley amidst the covid19 pandemic: Topic modeling using the latent dirichlet allocation (lda) approach. *SARIRIT: The University Research Journal*, *9*(1), 78–85.

Pezanowski, S., Mitra, N., & MacEachren, A. M. (2022). Exploring descriptions of movement through geovisual analytics. *arXiv preprint*. https://doi.org/10.1007/s42489-022-00098-3

Pressman, R. S., & Maxim, B. R. (2020). *Software engineering: A practitioner's approach* (9th). McGraw-Hill Education.

Rashidi, M., Ghodrat, M., Samali, B., & Mohammadi, M. (2018, October). Decision support systems. https://doi.org/10.5772/intechopen.79390

Republic of the Philippines. (2012). Republic act no. 10173: An act protecting individual personal information in information and communications systems in the government and the private sector, creating for this purpose a national privacy commission, and for other purposes (data privacy act of 2012). https://www.officialgazette.gov.ph/2012/08/15/republic-act-no-10173/

Rodolfo, K., Remootigue, C., & Lamug, C. (2003). Worsening floods around northern manila bay, philippines: Research-based analysis from physical and social science perspectives. *Philippine Sociological Review*, *51*, 17–40.

Royal HaskoningDHV. (2026). North Manila Bay Delta characteristics [Accessed: 2026-11-02]. https://northmanilabayfps.ireport.royalhaskoningdhv.com/updated-protection-strategy/root-cause--analysis/north-manila-bay-delta-characteristics

Singh, A. K. (2020). Chi-square test. https://www.lkouniv.ac.in/site/writereaddata/siteContent/20200426125814452 3Anoop_Applied_Chi_Square_Test.pdf

Sommerville, I. (2016). *Software engineering* (10th). Pearson Education Limited.

Stryker, C. (2021, October). *What are large language models (LLMs)?* IBM. Retrieved October 31, 2026, from https://www.ibm.com/think/topics/large-language-models

Svana, M., Zapletal, F., & Volny, J. (2026). Improving municipal decision-making with topic modeling and sentiment analysis. *Soft Computing*, *30*, 1617–1635. https://doi.org/10.1007/s00500-025-10976-3

United Nations. (2020, September). *SDG resource document: Targets overview* [Based on United Nations General Assembly Resolution A/RES/70/1]. https://sdgs.un.org/sites/default/files/2020-09/SDG%20Resource%20Document_Targets%20Overview.pdf

World Bank Group. (2023). *Philippines - country climate and development report: Background paper ph-11 disaster risk management in the philippines* (tech. rep.). World Bank. http://documents.worldbank.org/curated/en/099100523112042117

Xiao, R. (2010). Corpus creation. In *Handbook of natural language processing* (2nd, pp. 147–165). CRC Press. https://www.lancaster.ac.uk/fass/projects/corpus/ZJU/xpapers/Xiao_corpus_creation.pdf

Yang, A., et al. (2025). Qwen3 technical report.

Zang, Z. (2026). Analysis of public opinions on natural disasters on short-video social media based on the latent dirichlet allocation and prompt-bert-textcnn models. *Geoenvironmental Disasters*, *13*(1), Article 2626792. https://doi.org/10.1080/19475705.2026.2626792

System B Prompts

Listing 1: Stage 1 Prompt

```
                            TOPIC_DISCOVERY_SYSTEM ="""
                            You are an expert qualitative data
analyst. Your task is to analyze a batch of text documents and
synthesize a canonical taxonomy of exactly {n_topics} distinct
topics.

                            GUIDELINES:
                            - Focus on the core ACTION,
PROBLEM, or THEME (e.g., medical relief, structural damage,
supply distribution).
                            - Strictly omit all proper nouns,
locations, regions, and geographic identifiers.
                            - Each topic must have a short
'label' (2-4 lowercase words), and actual 'keywords'
(bag-of-words) extracted directly from the texts.
                            - Respond ONLY with a valid JSON
array of objects. No markdown, no explanations.
                            - CRITICAL: Your output MUST start
with "[" and end with "]".

                            EXAMPLE OUTPUT:
                            [
                            {{"label": "rescue efforts",
"keywords": ["rescue", "stranded", "volunteer", "bagal",
"rescuers"]}},
                            {{"label": "flood impact",
"keywords": ["baha", "tubig", "lubog", "taas", "stranded"]}},
                            {{"label": "health risks",
"keywords": ["leptospirosis", "sintomas", "check-up", "baho"]}}
                            ]
                            /no_think
                            """

                            TOPIC_DISCOVERY_USER = """
                            Here is a representative sample of
{n_docs} documents from the dataset:

                            {documents_block}

                            Generate the JSON array of exactly
{n_topics} topics based on the themes present in these
documents.
                            JSON Output:
                            """
```

Research Instrument

Human Assessment of Facebook Content and NLP Pipeline Outputs

Assessor ID: _____ **Date:** _____

Instructions. For each sampled Facebook content item, independently classify its sentiment. Then evaluate whether the topic representation generated by each system adequately describes the main subject of the content. The source type and identities of the two systems are concealed. Complete the human sentiment assessment before viewing the system outputs. Check the box of your answer.

Table 7: Human Assessment of Sampled Facebook Content

Sample ID	Sampled Facebook Content	Human Sentiment	System X: Topic Words	X Topic Alignment	System Y: Topic Words	Y Topic Alignment
001	[Sampled content]	□ Positive. □ Negative. □ Neutral.	word1, word2, word3, . . .	□ Yes □ No	word1, word2, word3, . . .	□ Yes □ No
002	[Sampled content]	□ Positive. □ Negative. □ Neutral.	word1, word2, word3, . . .	□ Yes □ No	word1, word2, word3, . . .	□ Yes □ No
003	[Sampled content]	□ Positive. □ Negative. □ Neutral.	word1, word2, word3, . . .	□ Yes □ No	word1, word2, word3, . . .	□ Yes □ No

*Note.*Topic alignment is marked Yes if the topic representation adequately describes the main subject of the sampled content, and No otherwise.

Listing 2: Stage 2 Prompt

```
                            JOINT_SYSTEM = """
                            You are an expert analyst
classifying social media posts. You must output TWO labels:
SENTIMENT and TOPIC.

                            SENTIMENT must be exactly one of:
"POS", "NEG", "NEU".
                            - POS: The text expresses joy,
gratitude, hope, relief, or praise.
                            - NEG: The text expresses anger,
sadness, fear, frustration, or criticism.
                            - NEU: The text is purely
informational, factual, or expresses a balanced/mixed view. Do
not assume factual requests are negative.

                            TOPIC must be exactly one of the
'label' values from the provided taxonomy list.

                            Respond ONLY with a valid JSON
object. No markdown, no explanations.
                            FORMAT:
                            {{"sentiment": "LABEL", "topic":
"exact topic label from list"}}
                            /no_think
                            """


                            JOINT_USER = """Available taxonomy
(label and description): {topic_list_with_descriptions}

                            Examples:
                            Text: "Salamat sa mga rescuers!
Ligtas na kami."
                            JSON: {{"sentiment": "POS",
"topic": "rescue efforts"}}

                            Text: "Base sa PAGASA, tuloy-tuloy
ang ulan hanggang weekend. Maghanda po."
                            JSON: {{"sentiment": "NEU",
"topic": "flood warning"}}

                            Text: "Nakakagalit! Ang bagal ng
rescue. Ilang oras na kaming stranded."
                            JSON: {{"sentiment": "NEG",
"topic": "flood impact"}}

                            Text to classify:"{text}"

                            JSON Response:
                            "69
```